

IDS Milestone-1

Project Overview

This project aims to develop an interactive dashboard for analyzing the Brazilian E-Commerce Public Dataset by Olist.

The tool will integrate multiple datasets to provide insights into product sales over the Brazilian States and help merchants make informed decisions.

Data Sources

Brazilian E-Commerce Public Dataset by Olist (Kaggle)

- Size: 126.19 MB
- Files:

olist_customers_dataset.csv

Attributes	Type
customer_id	Foreign key from orders to customers
customer_unique_id	Unique identifier (Primary key)
customer_zip_code_prefix	int
customer_city	string
customer_state	string

olist_geolocation_dataset.csv

Attributes	Type
geolocation_zip_code_prefix	int
geolocation_lat	float
geolocation_lng	float

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geolocation_city	string
geolocation_state	string

olist_order_items_dataset.csv

Attributes	Type
order_id	Unique identifier
order_item_id	int
product_id	Foreign key to products
seller_id	Foreign key to sellers
shipping_limit_date	Date type
price	float
freight_value	float

olist_order_payments_dataset.csv

Attributes	Type
order_id	Unique identifier
payment_sequential	int
payment_type	string
payment_installments	int
pyment_value	float

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olist_order_reviews_dataset.csv

Attributes	Type
review_id	Unique Identifier
order_id	Foreign key to Orders dataset
review_score	int
review_comment_title	string
review_creation_date	Date type
review_comment_message	string
review_answer_timestamp	timestamp

olist_orders_dataset.csv

Attributes	Type
order_id	Unique Identifier
customer_id	Foreign key to Customers
order_status	string
order_purchase_timestamp	timestamp
order_approved_at	timestamp
order_delivered_carrier_date	Date type
order_delivered_customer_date	Date type
order_estimated_delivery_date	Date type

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olist_products_dataset.csv

Attributes	Type
product_id	Unique identifier
product_category_name	string
product_name_lenght	float
product_description_lenght	float
product_photos_qty	float
product_weight_g	float
product_length_cm	float
product_height_cm	float
product_width_cm	float

olist_sellers_dataset.csv

Attributes	Type
seller_id	Unique identifier
seller_zip_code_prefix	int
seller_city	string
seller_state	string

product_category_name_translation.csv

Attributes	Type
product_category_name	string
product_category_name_english	string

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Technical Stack

Primary Technologies:

- Language: Python 3
- Data Storage: pandas dataframes
- Data Processing: Pandas, NumPy
- Visualization: Plotly, Seaborn, Matplotlib

Data Cleaning

Null Values Elimination

Geolocations Cleaning Wrong Addresses

- Grouped all the locations by zipcode in Geolocations to get all the valid, unique {zip code, city, state} combinations, stored them in a Python dict and checked all the customers' locations with these values to verify whether the addresses are valid. **Caught “7523” invalid entries.**

Orders

Had the below-mentioned number of null values for each of the following rows:

order_approved_at 160
order_delivered_carrier_date 1783
order_delivered_customer_date 2965

Removed the:

- rows with order_approved_at == NULL if the order_state is set to any state in the list: ['approved', 'processing', 'shipped', 'invoiced', 'delivered']
- rows with order_delivered_carrier_date == NULL if the order_state is set to any state in the list: ['invoiced', 'delivered']
- rows with order_delivered_customer_date == NULL if the order_state is set to any state in the list: ['delivered']

Products

- Cleaned the Missing Rows:
 - product_category_name 610
 - product_name_lenght 610
 - product_description_lenght 610
 - product_photos_qty 610
 - product_weight_g 2
 - product_length_cm 2
 - product_height_cm 2

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- product_width_cm 2
- The conditions:

```
df_products =  
df_products.drop(df_products[df_products['product_category_name'  
''].isnull() |  
  
((df_products['product_name_length'].isnull() |  
df_products['product_photos_qty'].isnull()) &  
  
df_products['product_description_length'].isnull()) |  
  
df_products['product_weight_g'].isnull() |  
  
(df_products['product_length_cm'].isnull() &  
  
df_products['product_height_cm'].isnull() &  
  
df_products['product_width_cm'].isnull())  
].index)
```

- Replaced all the Espanol values of 'product_name_category' in the df_products to English using the df_product_category_name_translation.

Additions

Reviews

- Added new attribute 'review_type': [good, neutral, bad]

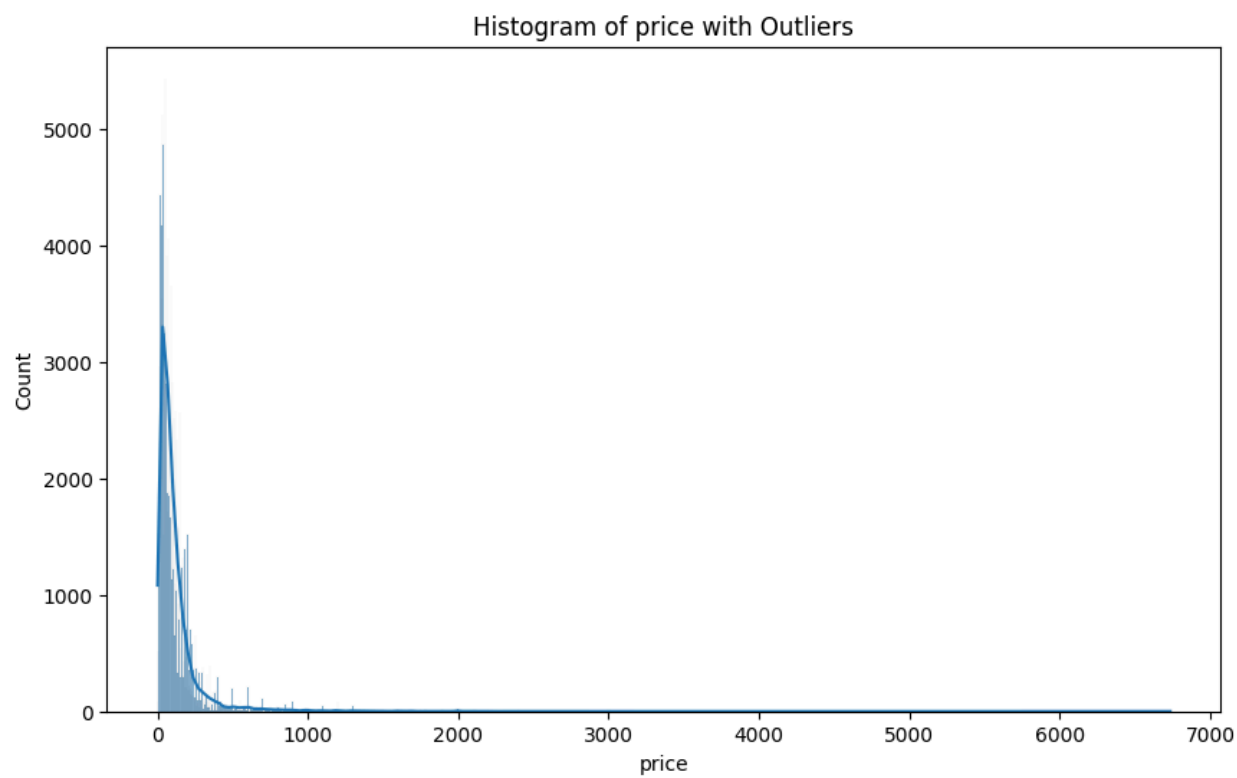
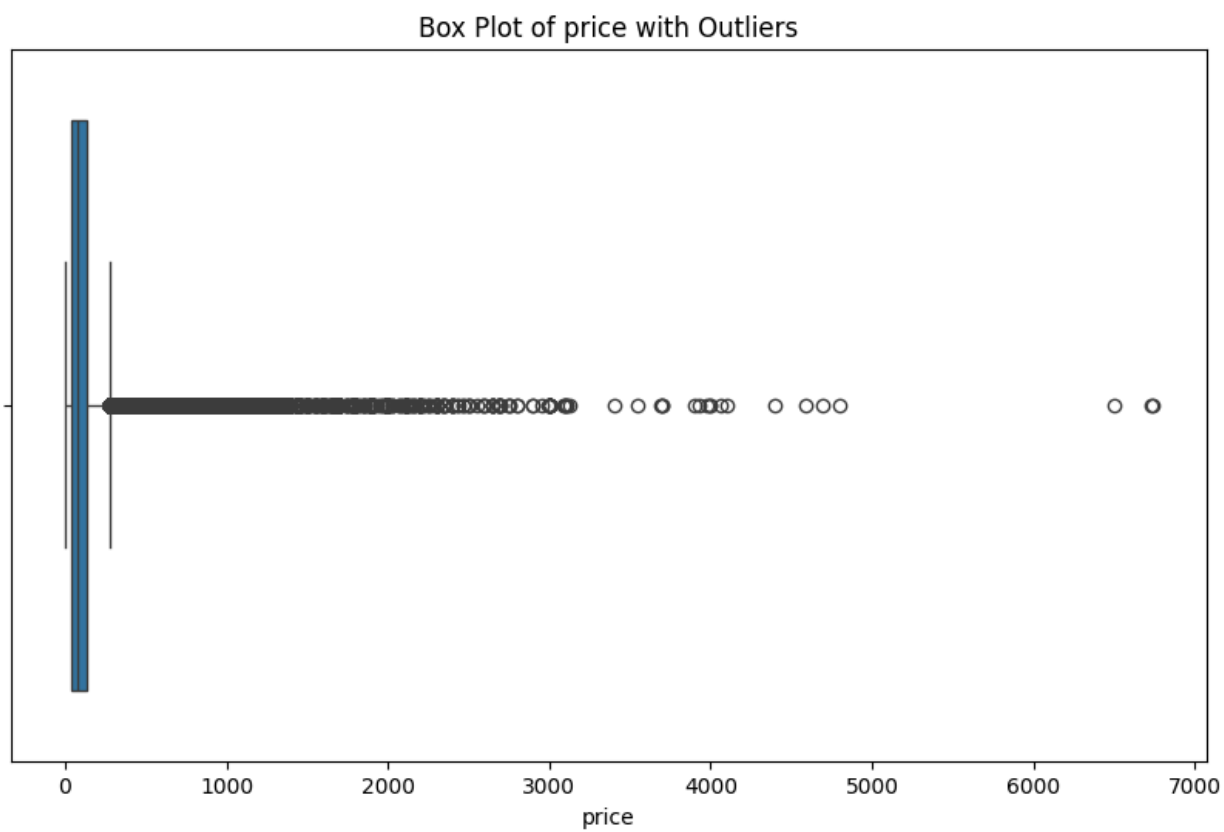
Outlier Analysis

Order Items Table

Price

- Number of outliers (IQR): 8427
- Number of outliers (Z-score): 1966

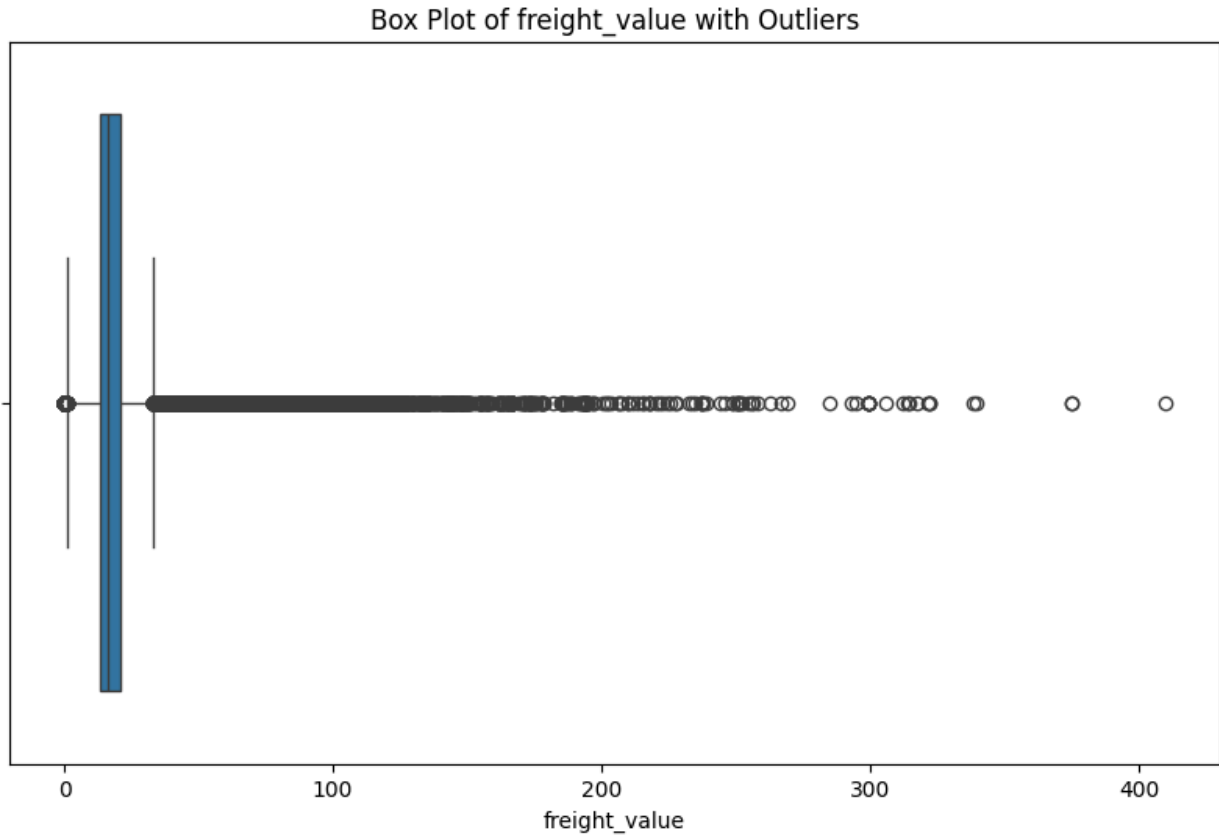
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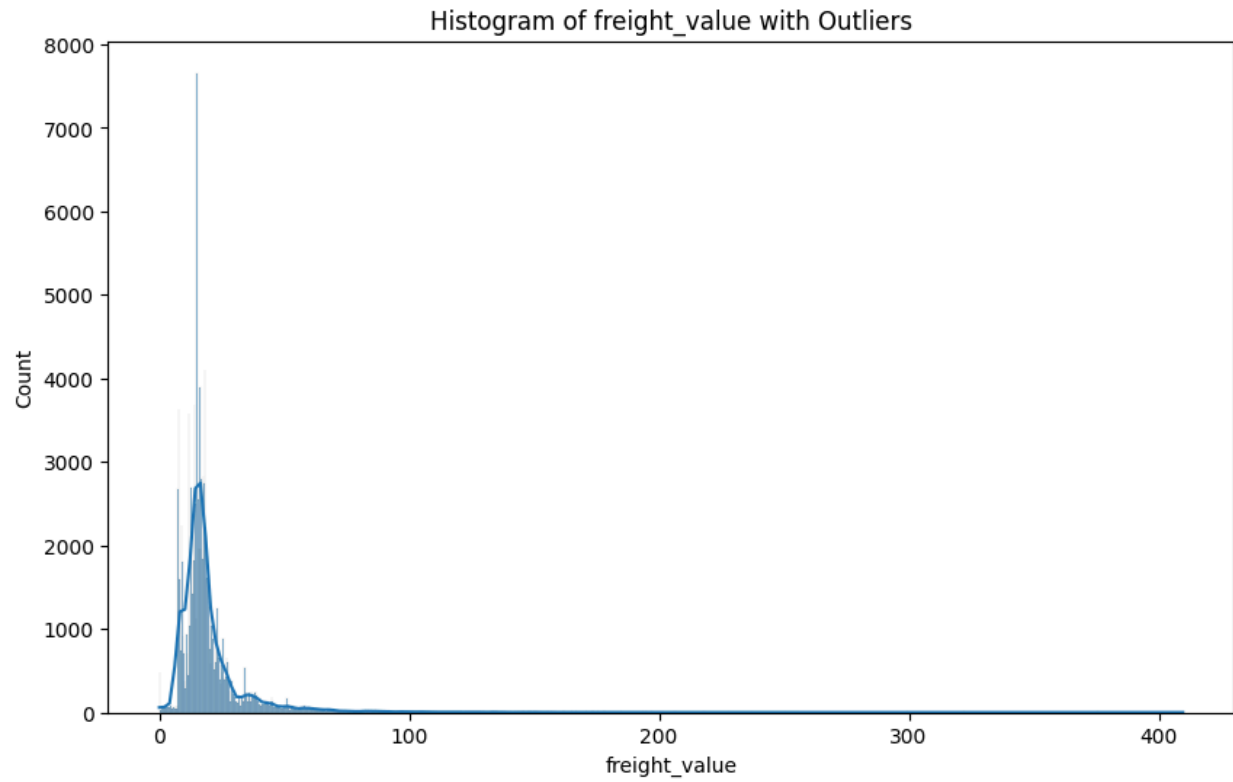
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Freight Value

- Number of outliers (IQR): 12134
- Number of outliers (Z-score): 2041



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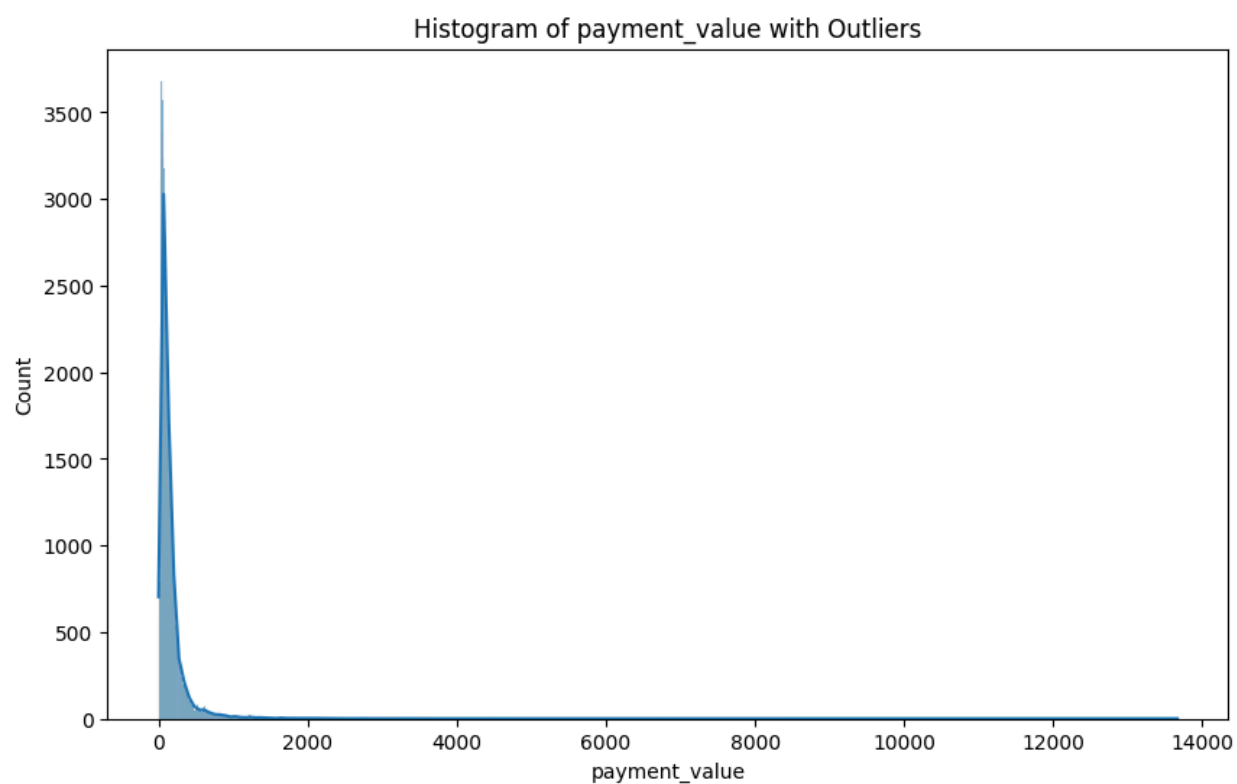
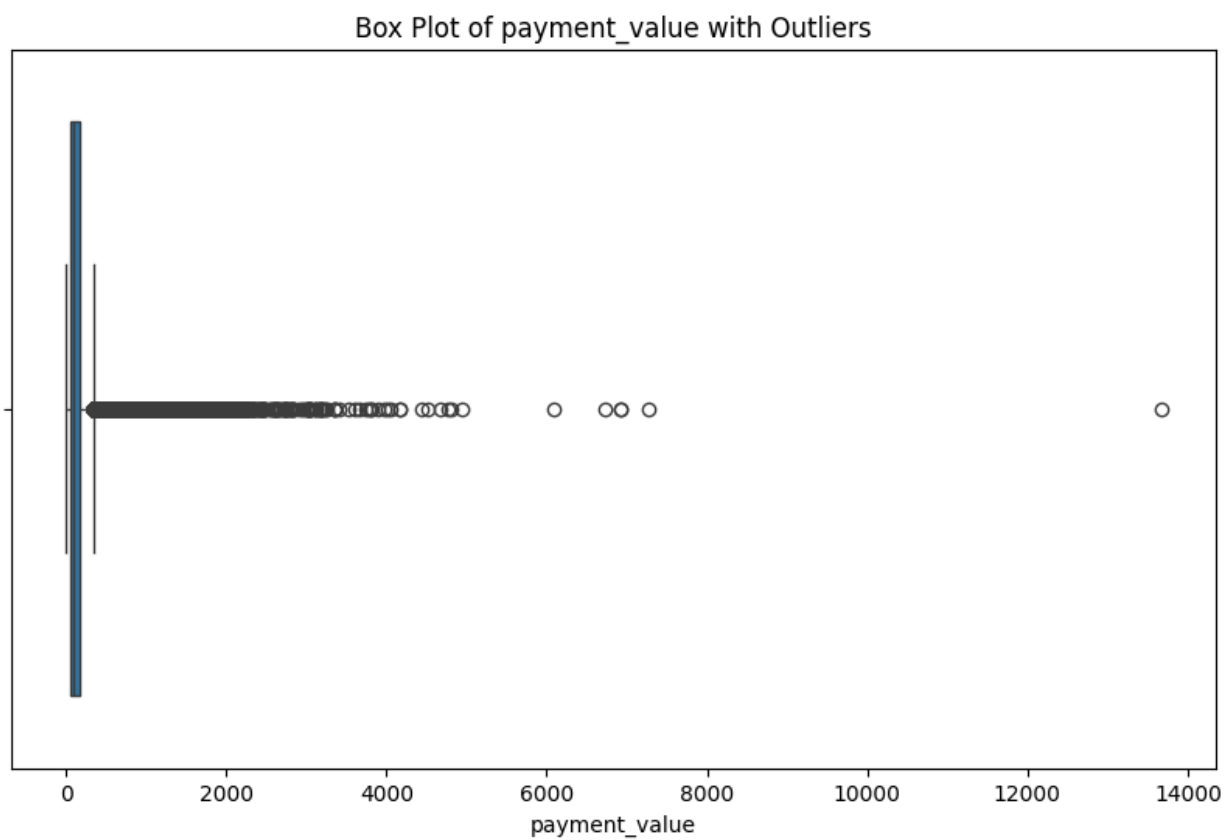


Payments Table

Payment Value

- Number of outliers (IQR): 7981
- Number of outliers (Z-score): 1803

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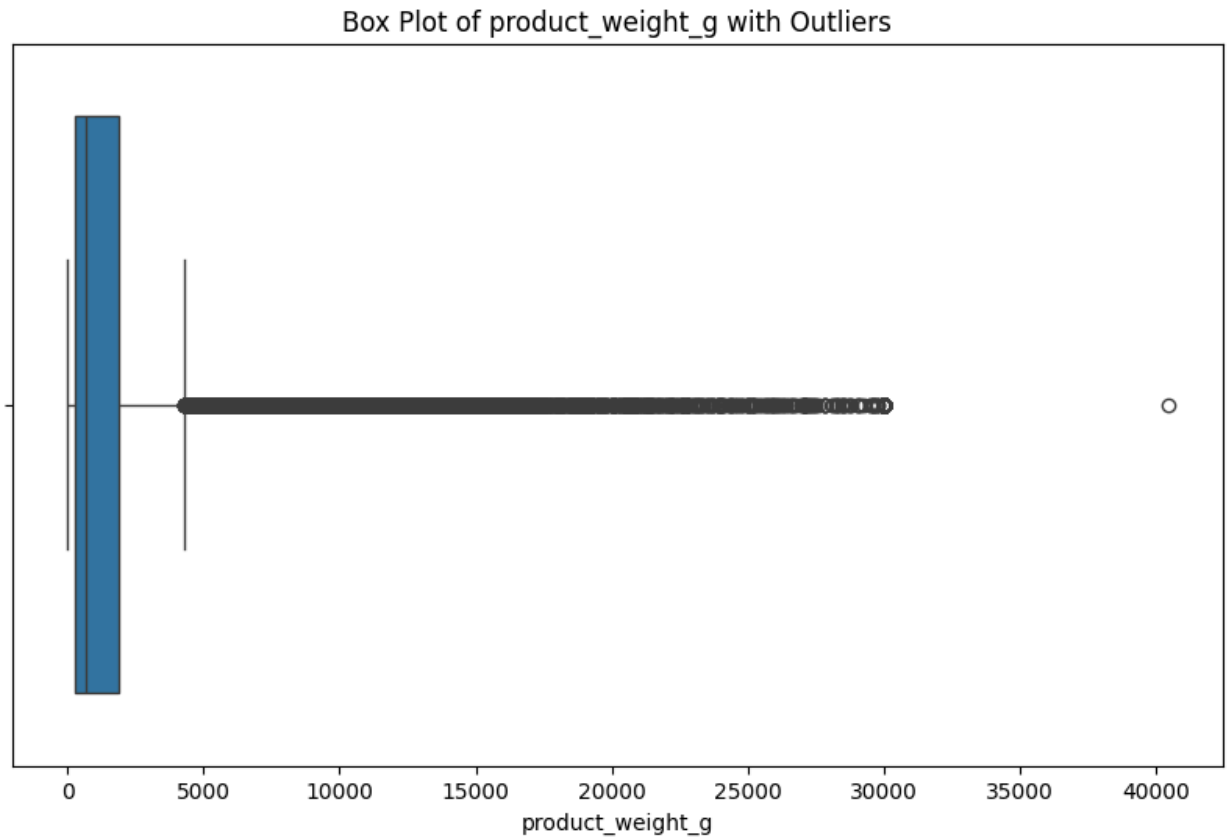


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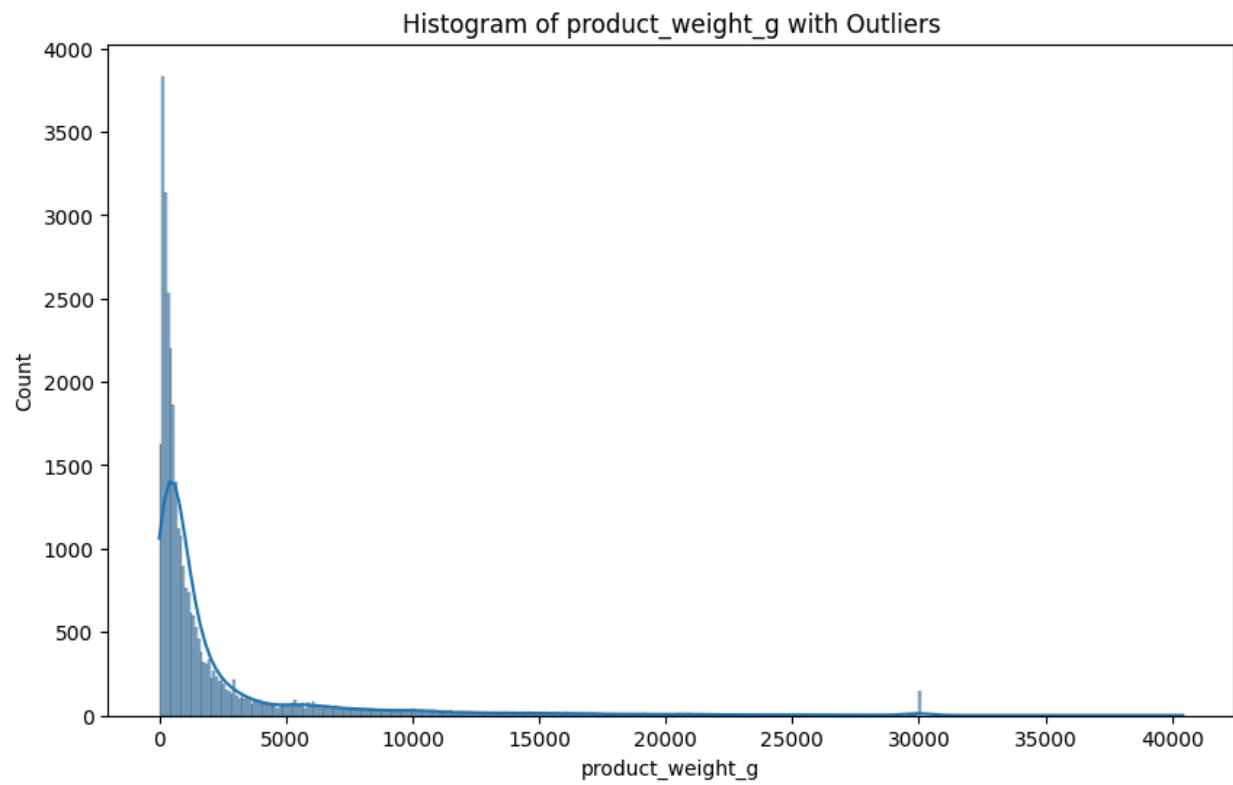
Products Table

Product weight (g):

- Number of outliers (IQR): 4478
- Number of outliers (Z-score): 937



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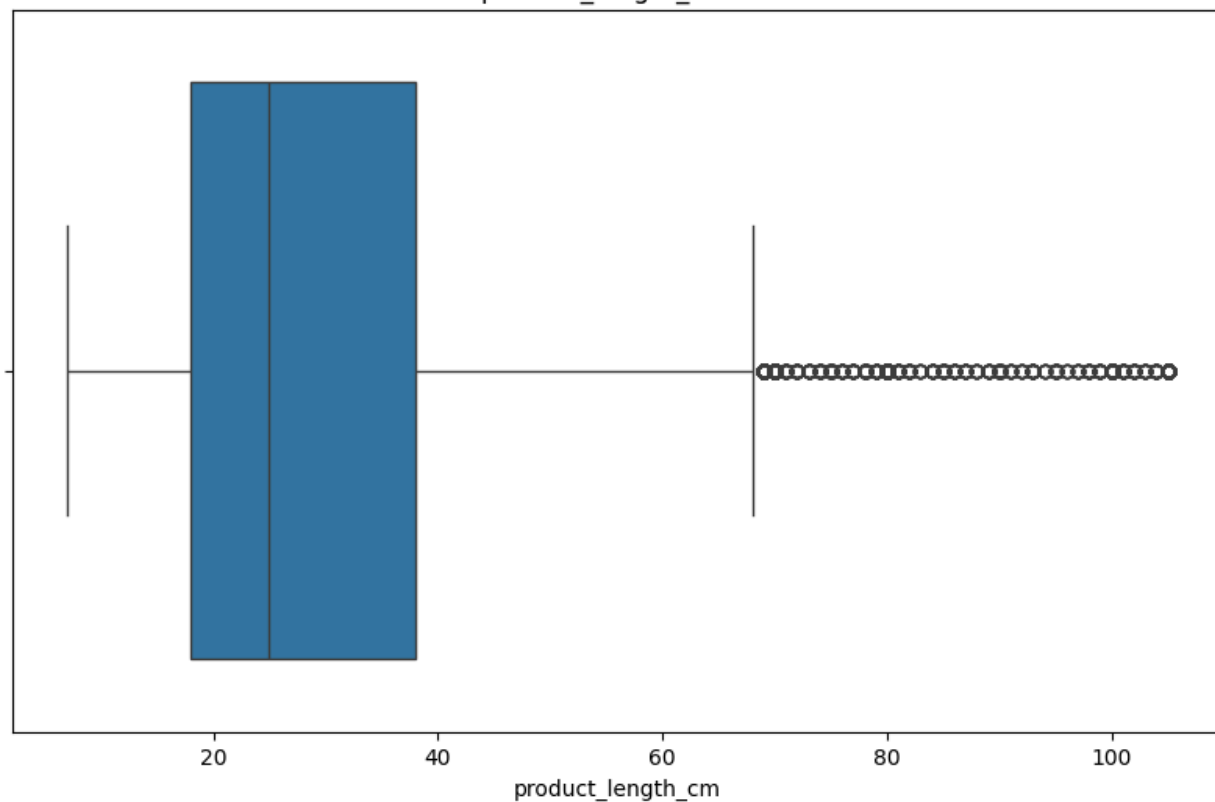


Product length (cm)

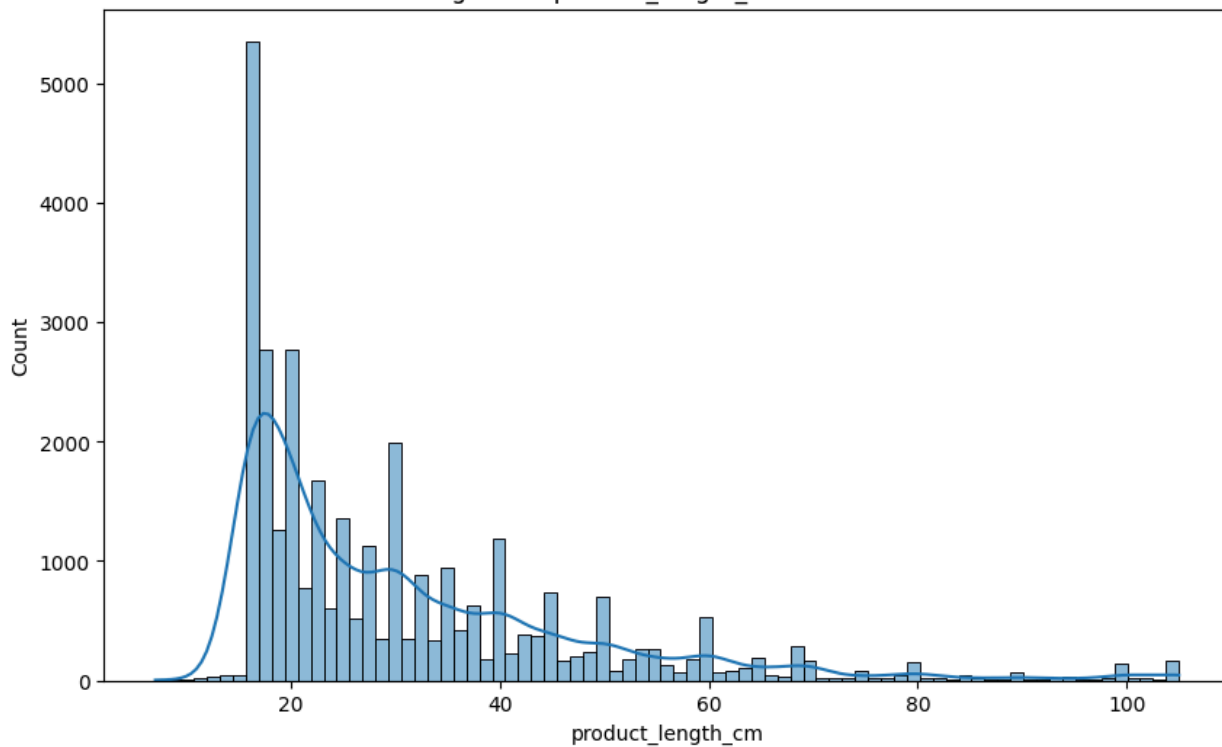
- Number of outliers (IQR): 1365
- Number of outliers (Z-score): 613

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Box Plot of product_length_cm with Outliers



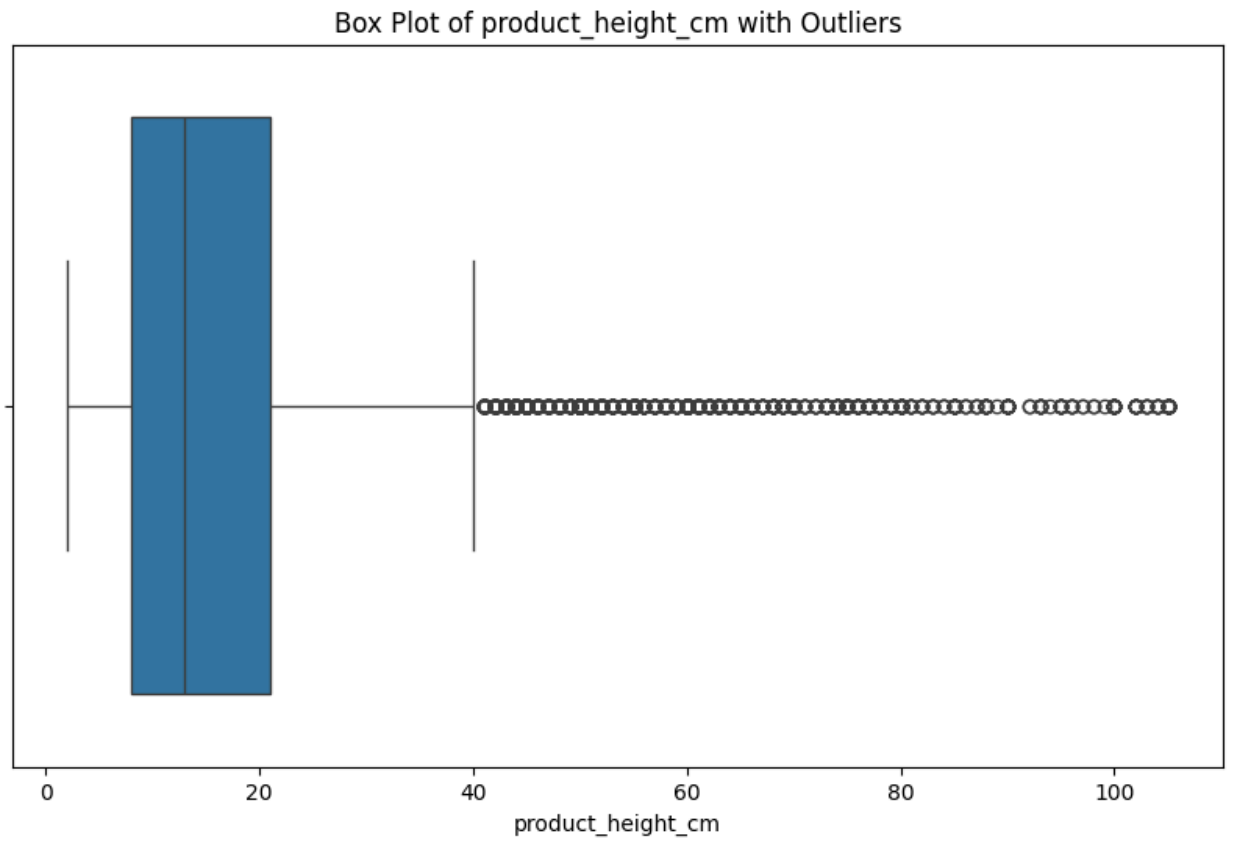
Histogram of product_length_cm with Outliers



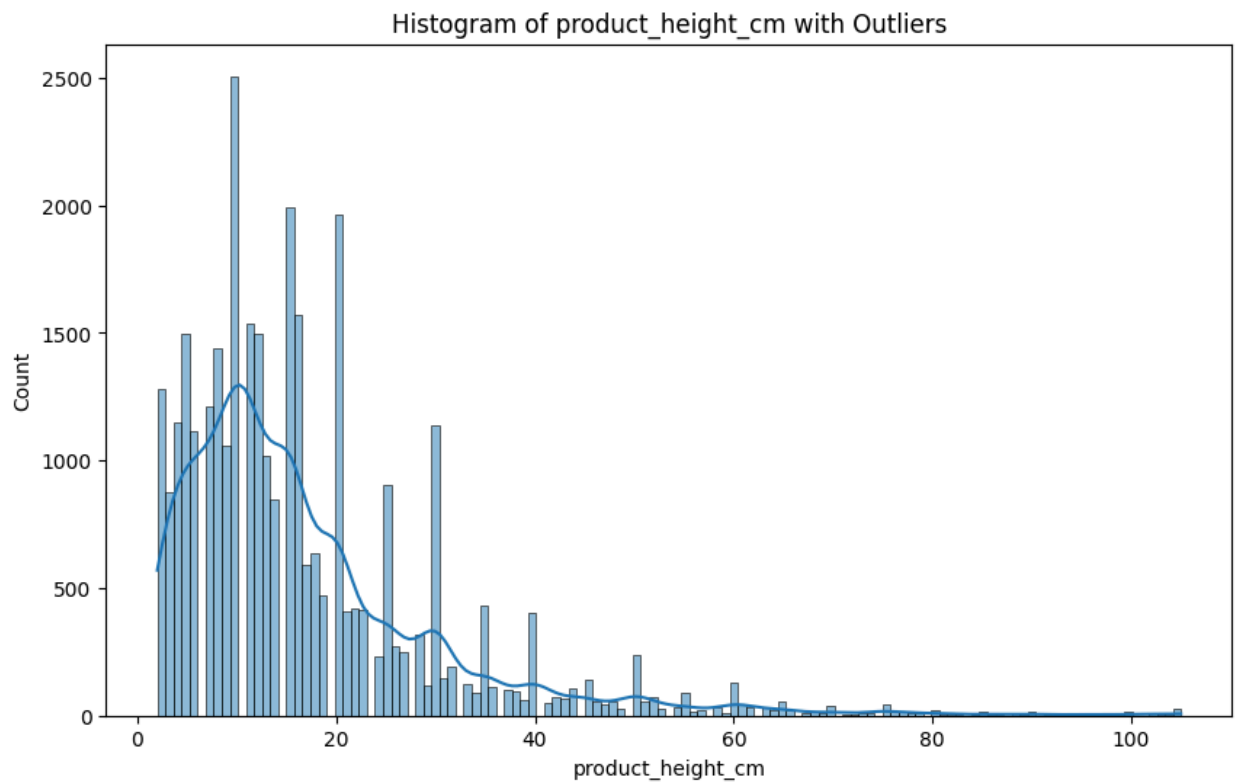
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Product height (cm)

- Number of outliers (IQR): 1870
- Number of outliers (Z-score): 719



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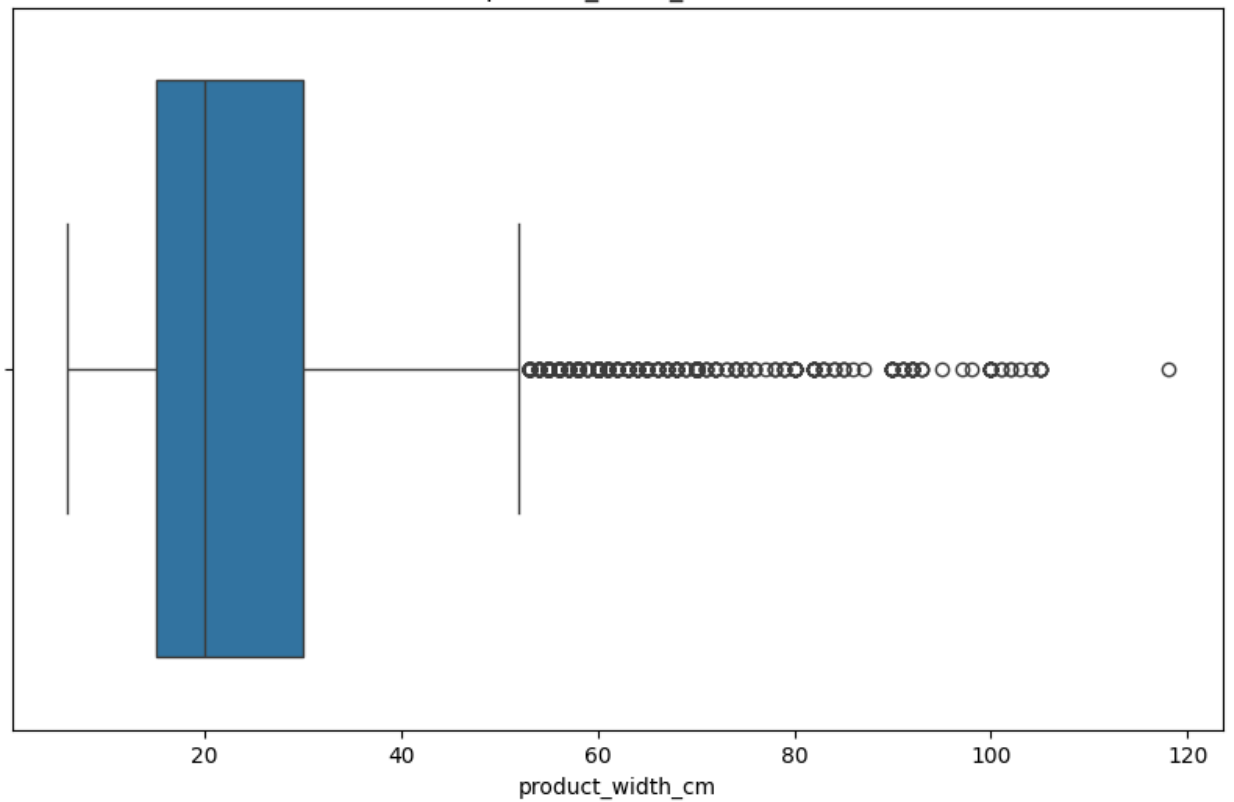


Product width cm

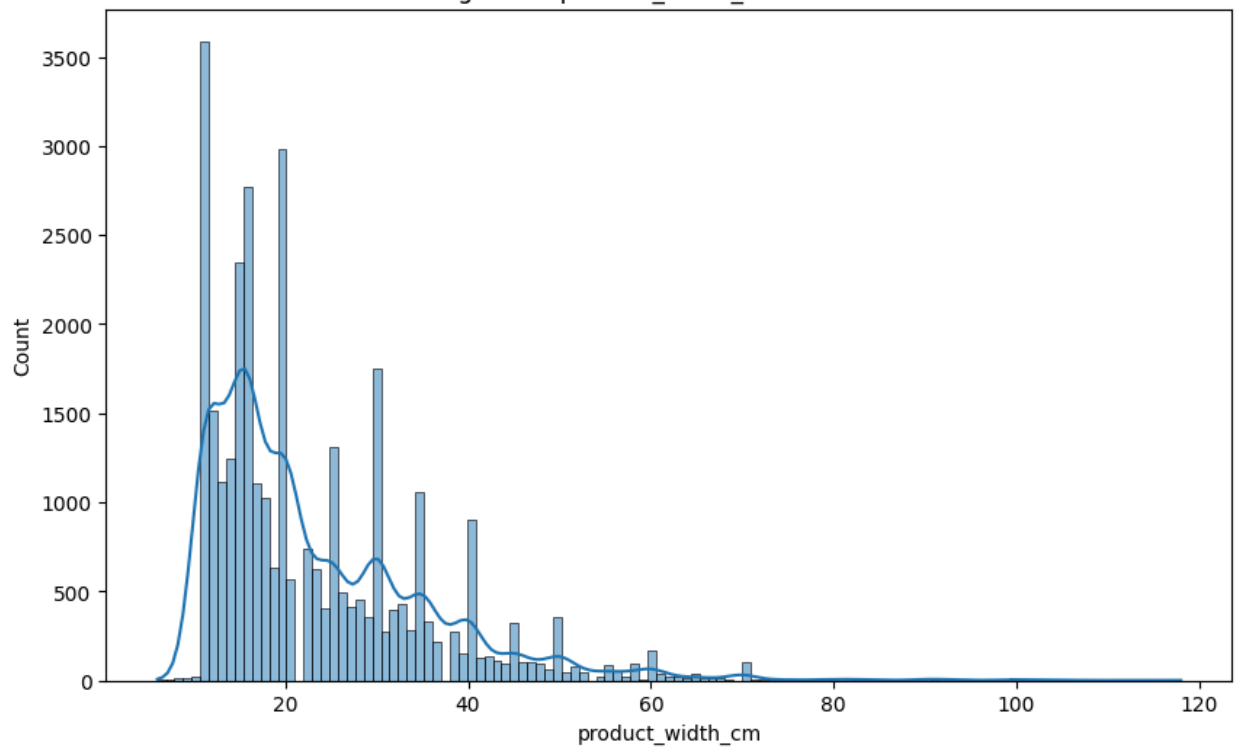
- Number of outliers (IQR): 897
- Number of outliers (Z-score): 567

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Box Plot of product_width_cm with Outliers



Histogram of product_width_cm with Outliers

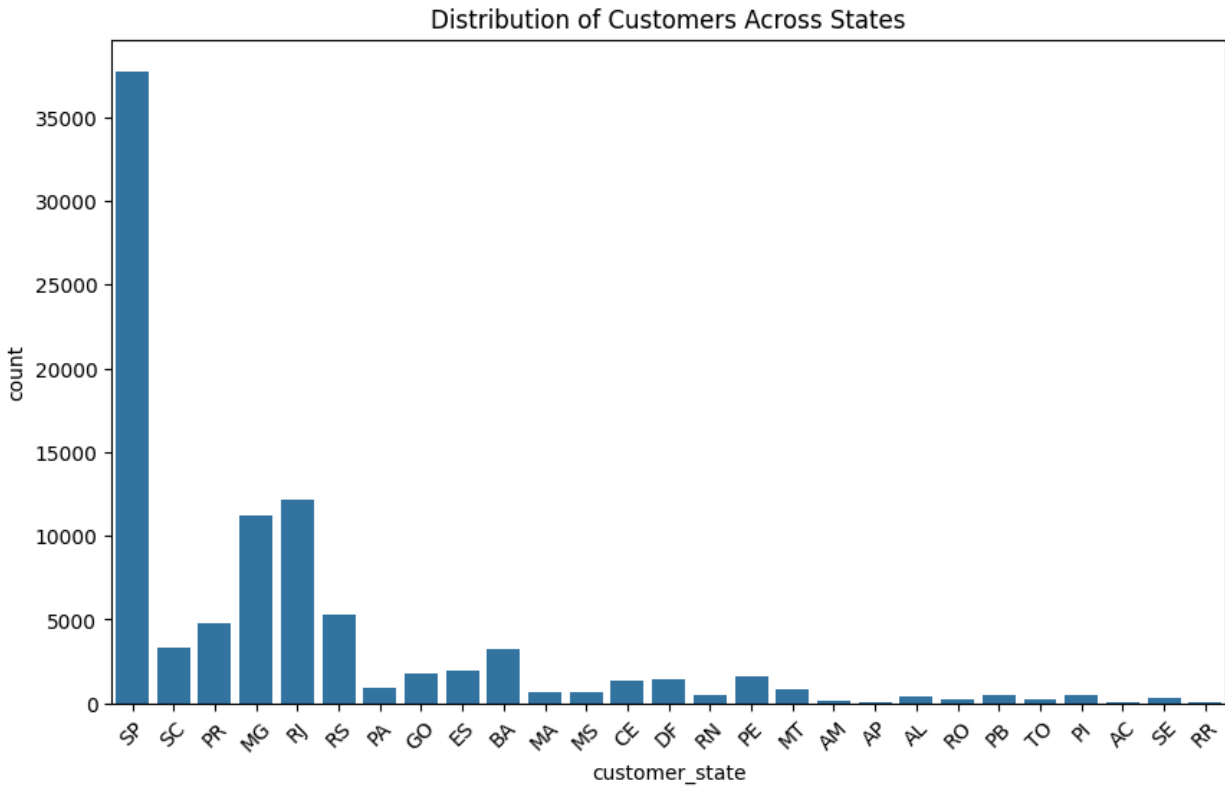


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Descriptive Analysis

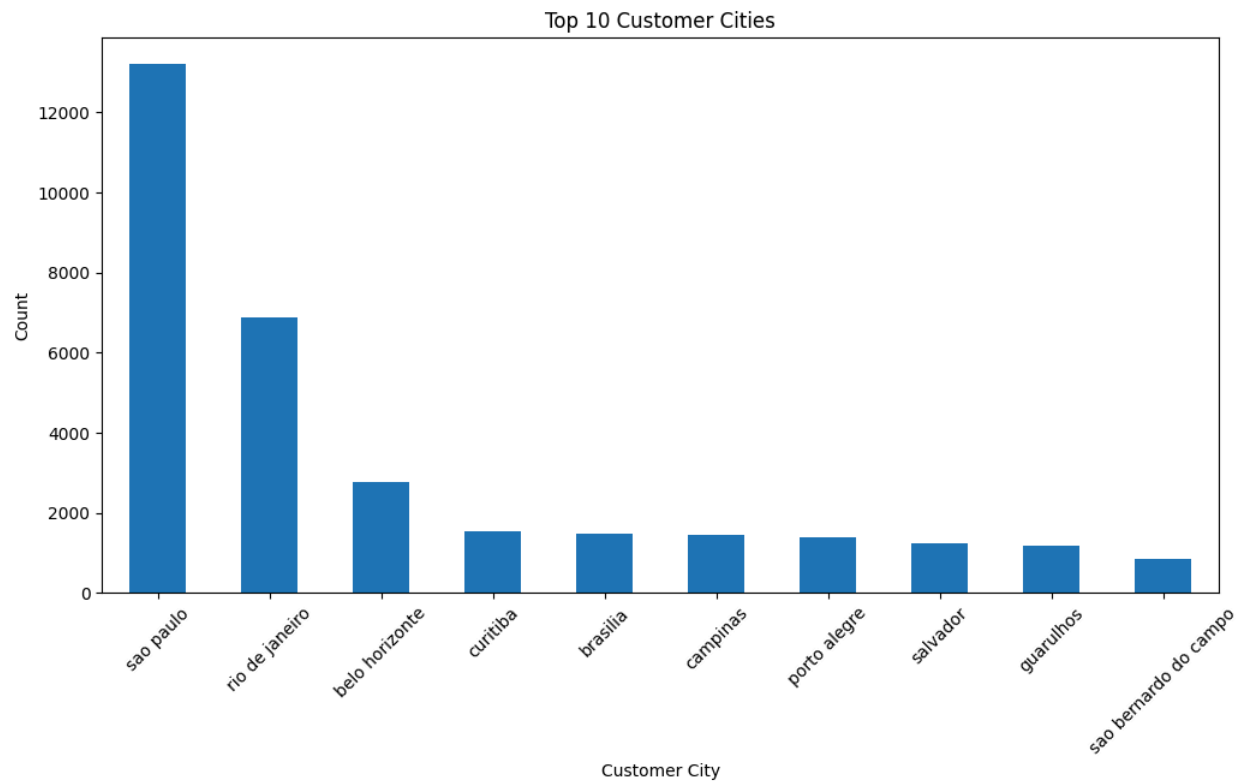
Customers Table

Distribution of Customers Across the State



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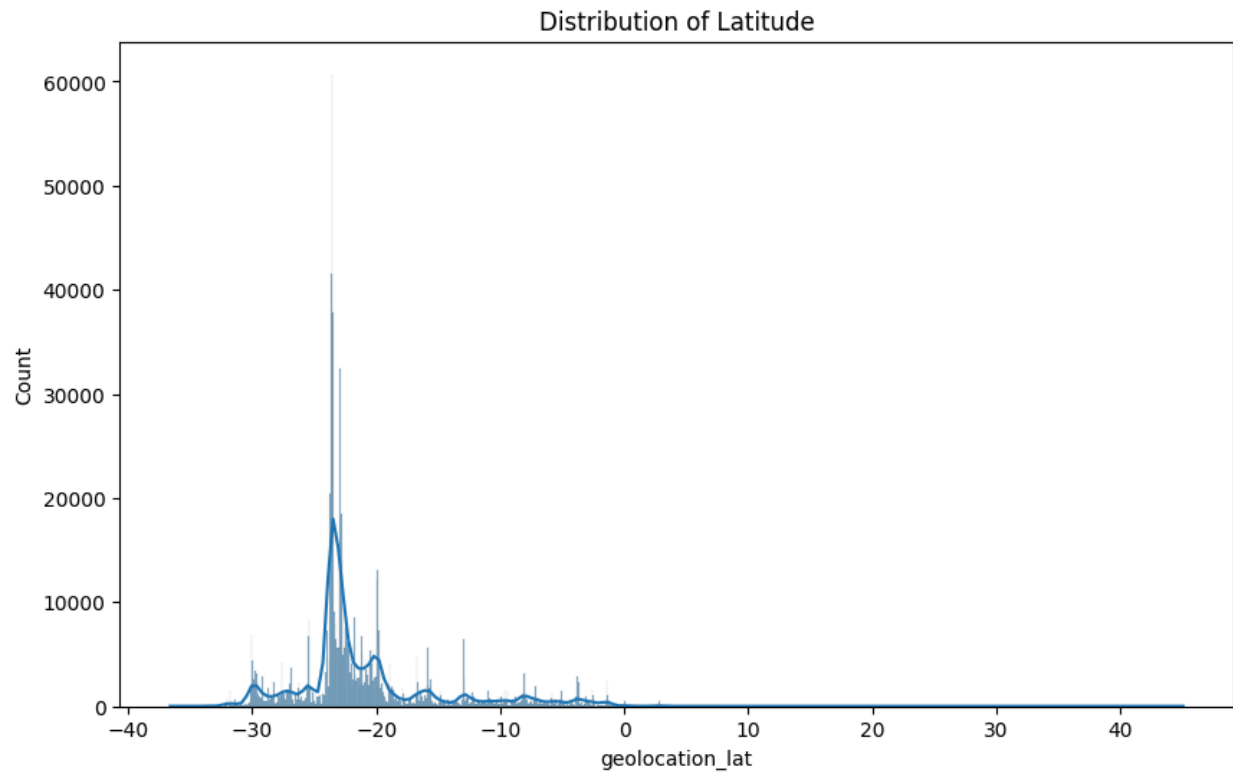
Top 10 Customer Cities



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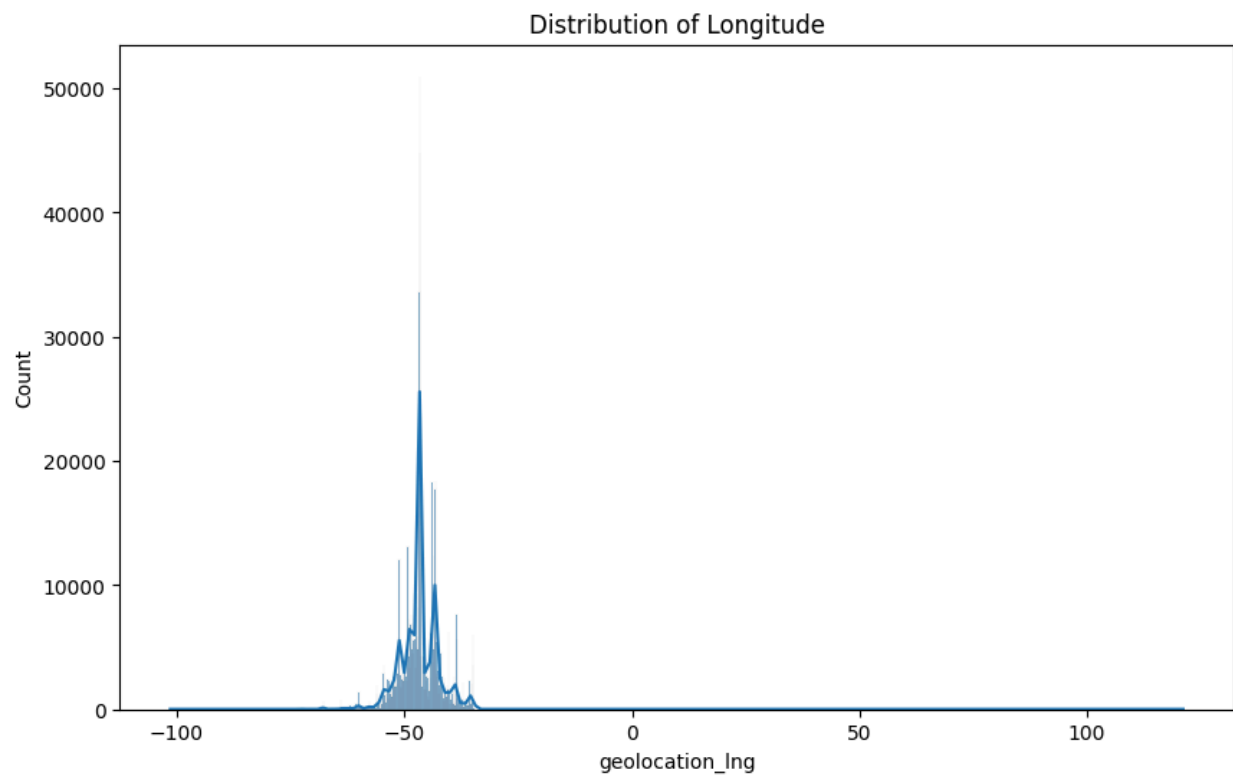
Geolocations Table

Distribution of Latitude



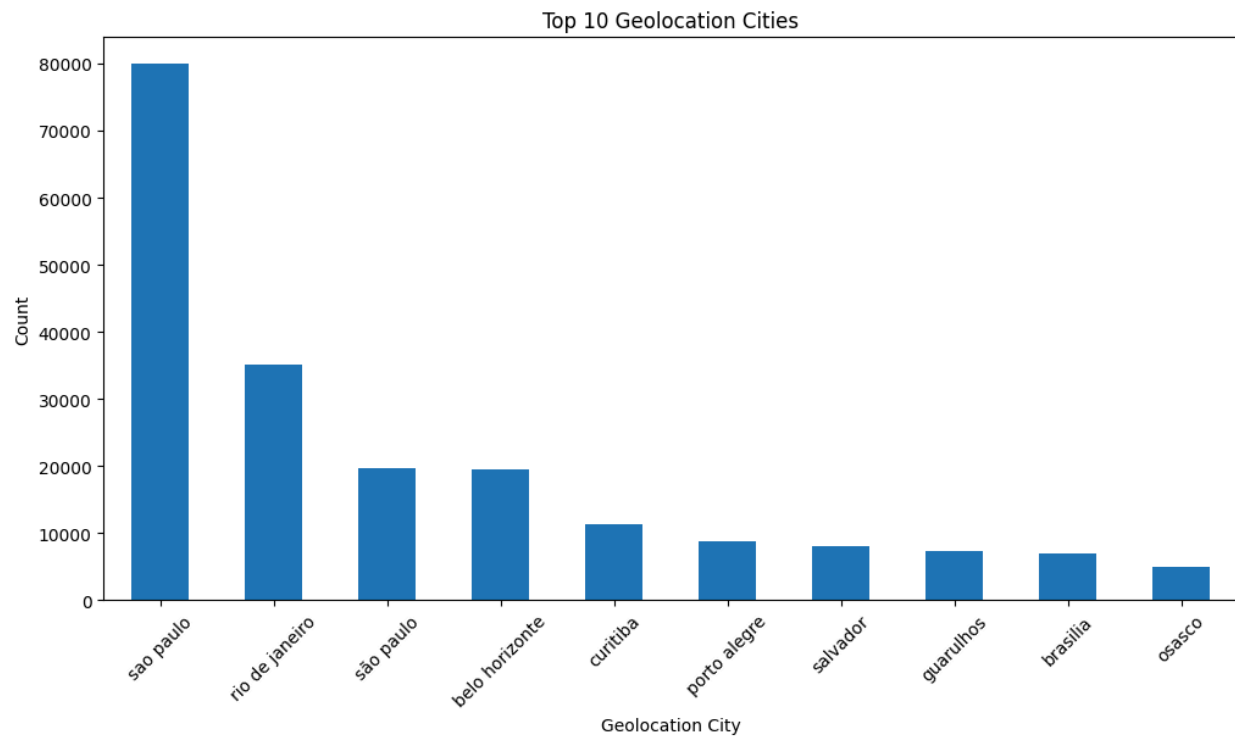
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Distribution of Longitude



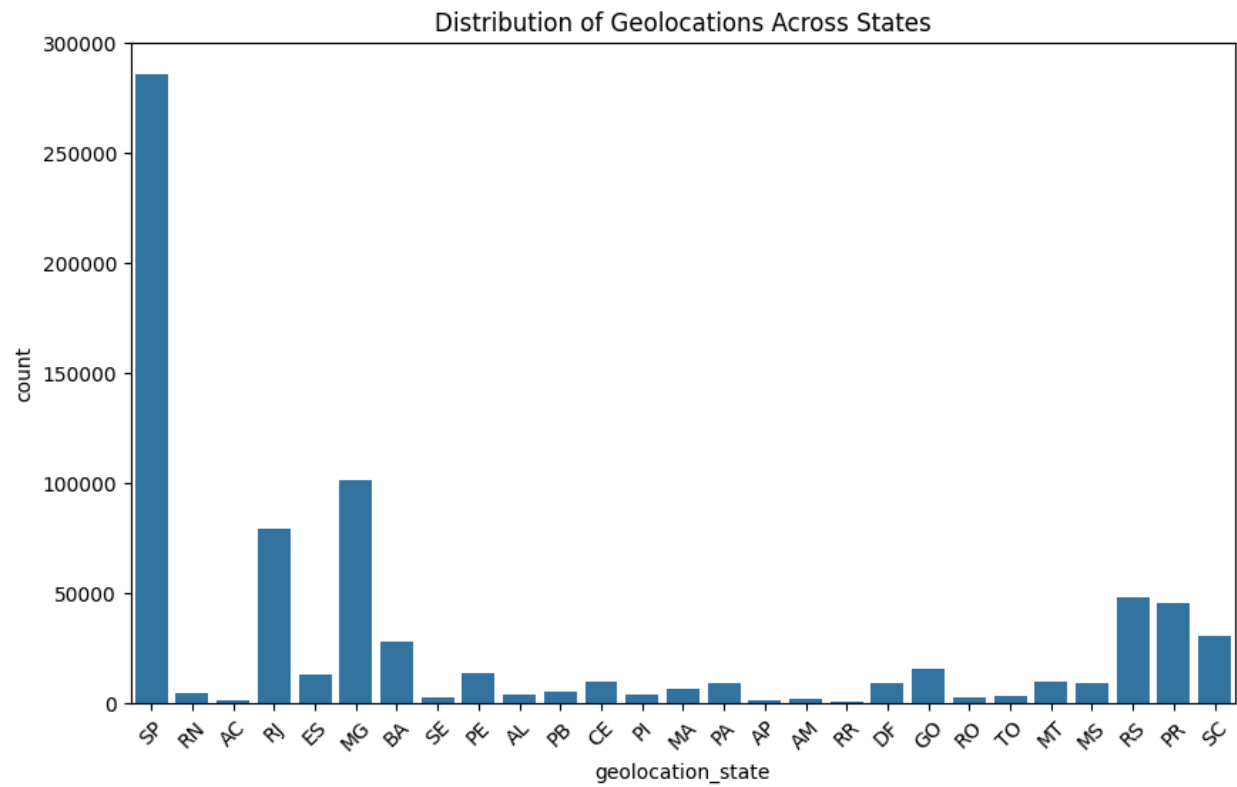
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Top 10 Geolocation Cities



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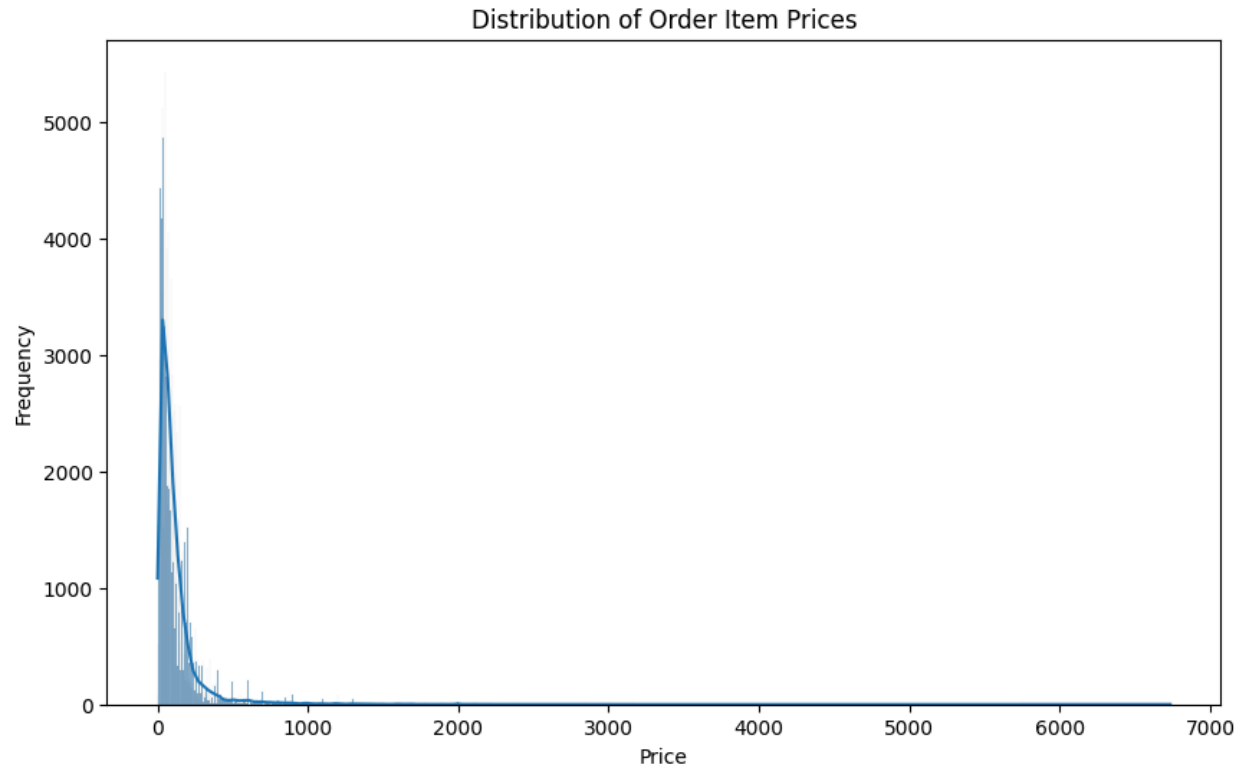
Distribution of Geolocations across the States



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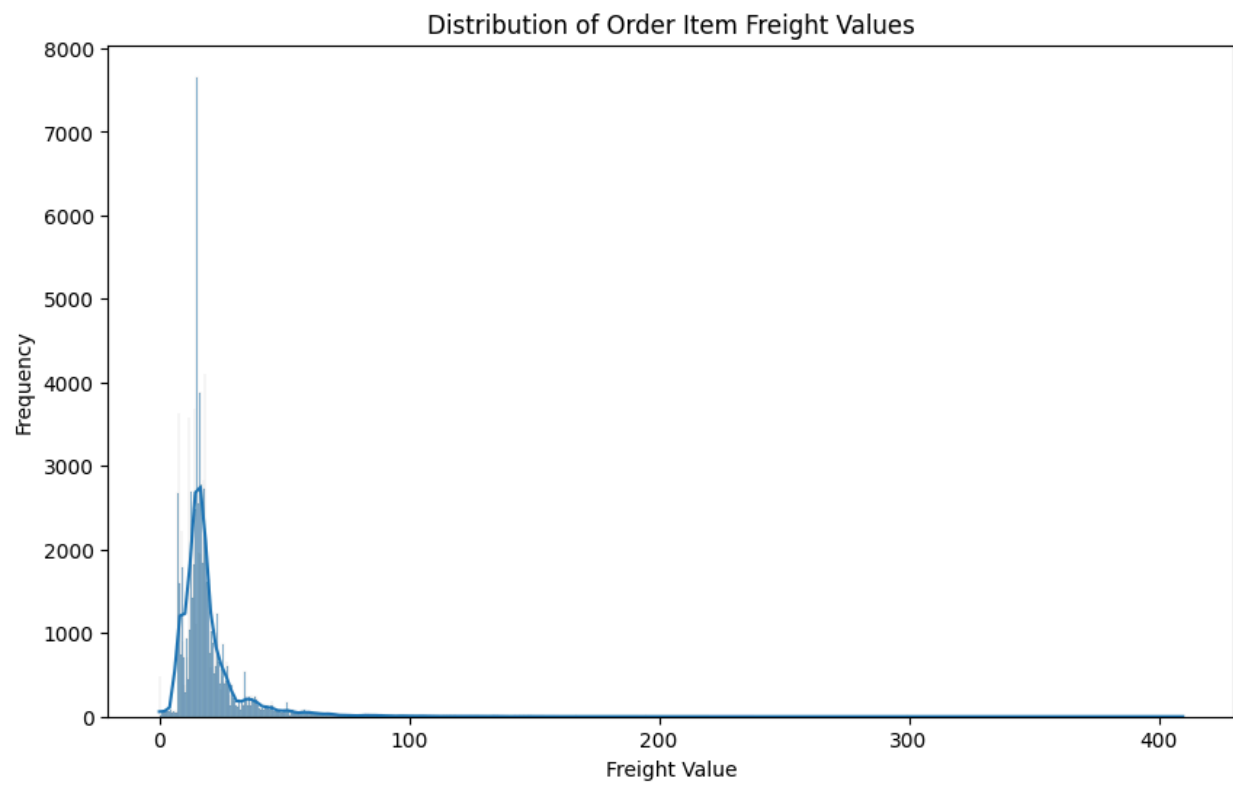
Order Items Table

Distribution of Order Item Prices



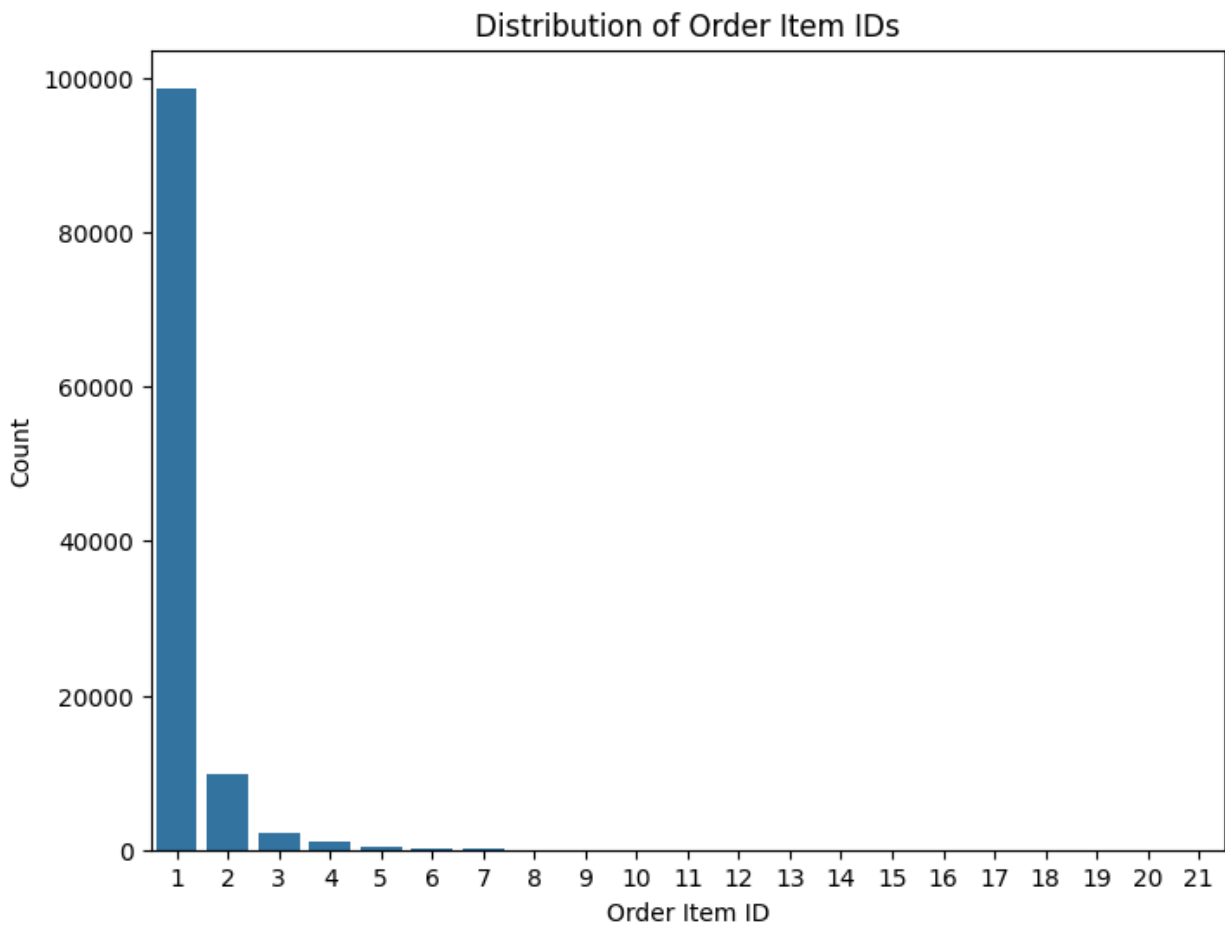
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Distribution of Order Item Freight Values



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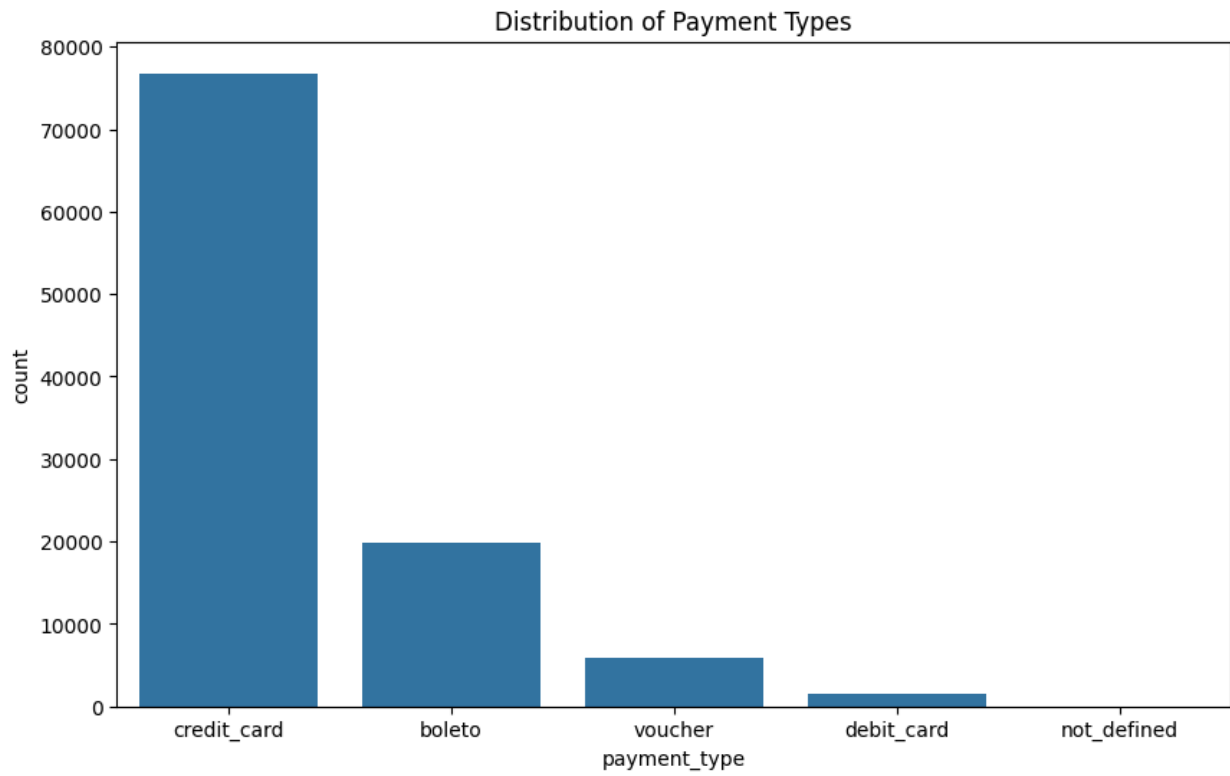
Distribution of Order Item IDs



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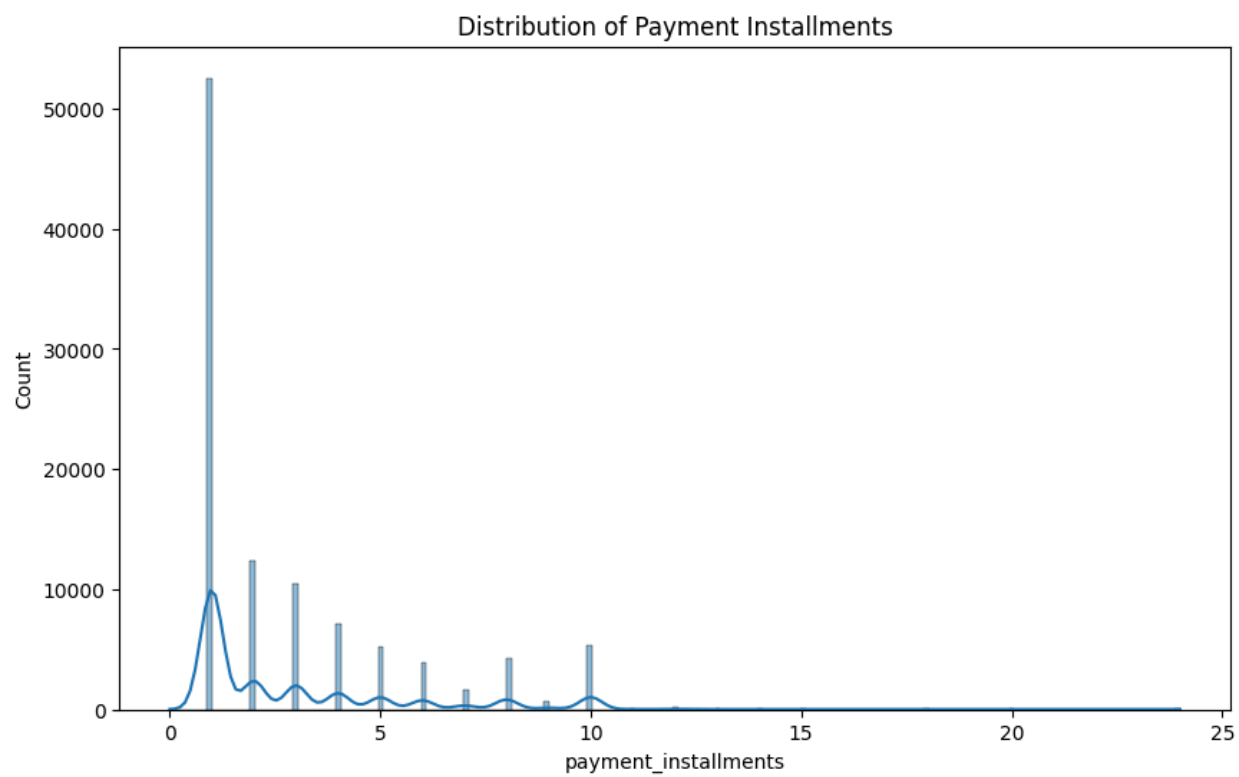
Order Payments Table

Distribution of Payment types



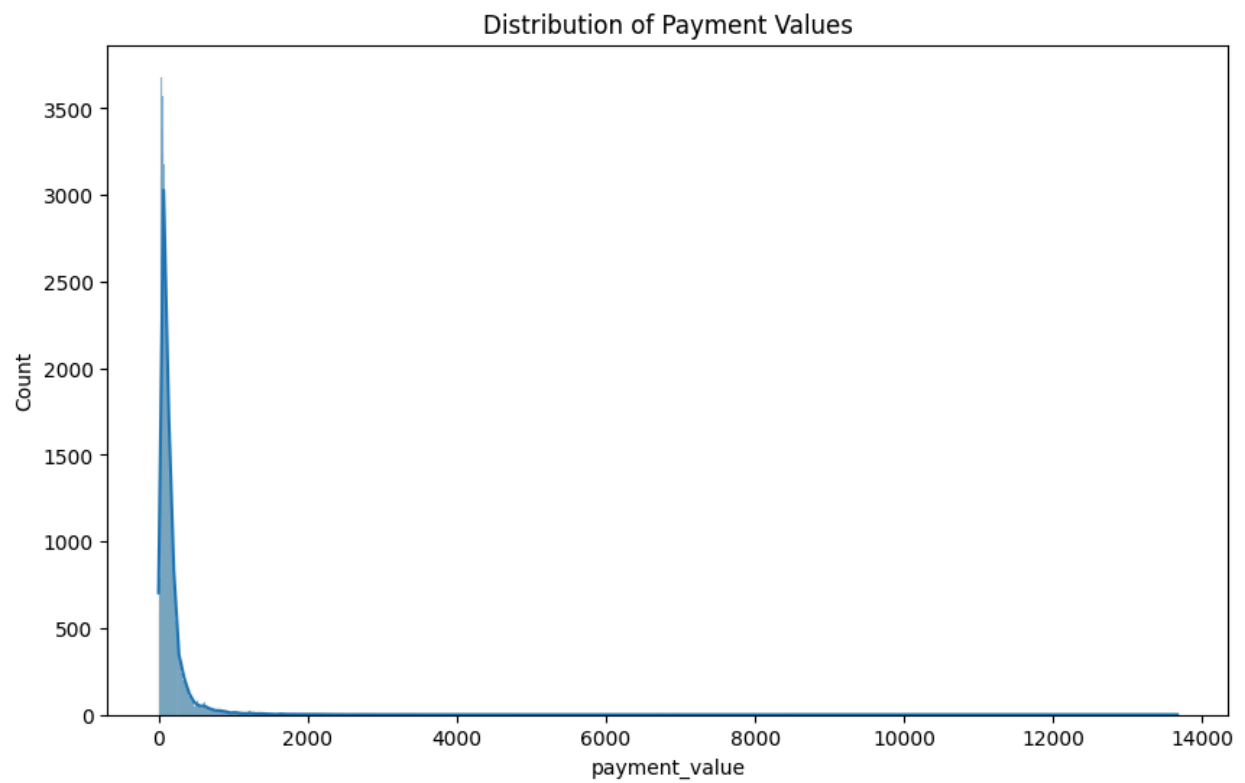
Jaiharishan Arunagiri Veerakumar (62333614)
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Distribution of Payment Installments



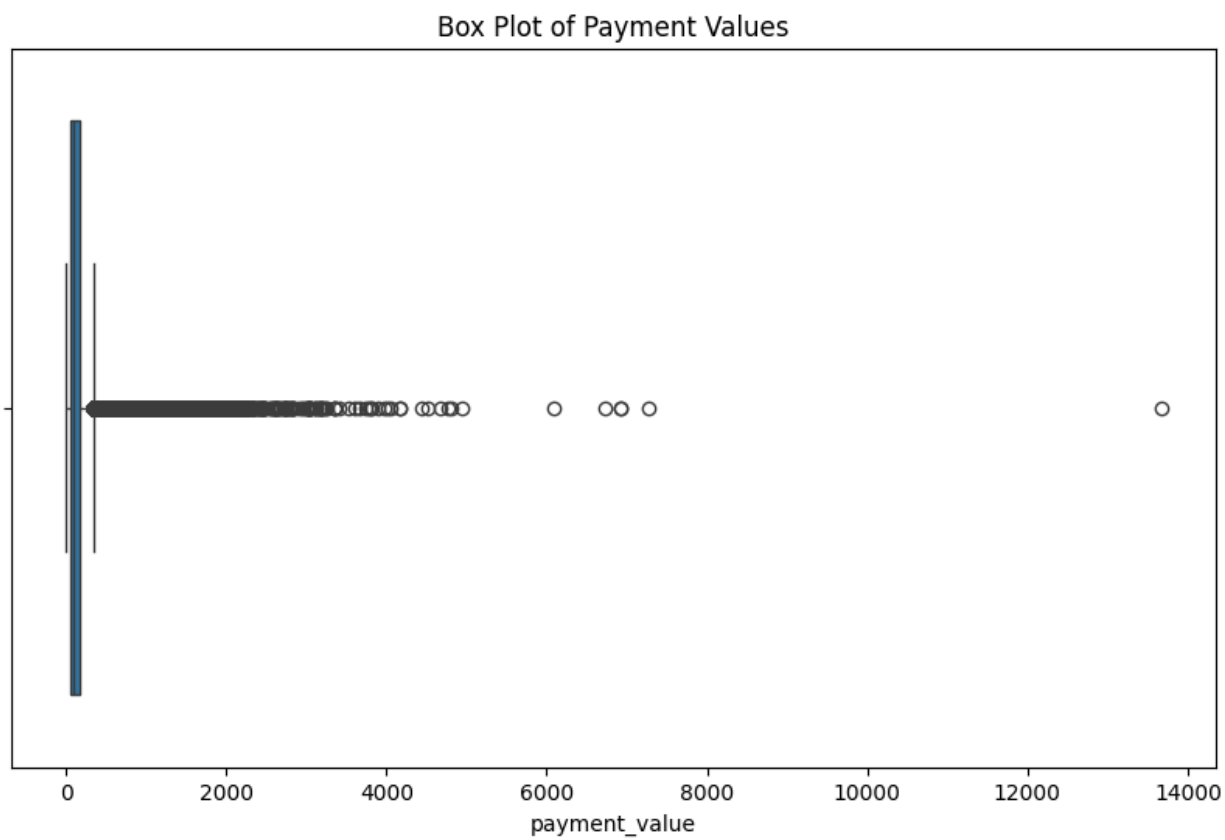
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Distribution of Payment Values



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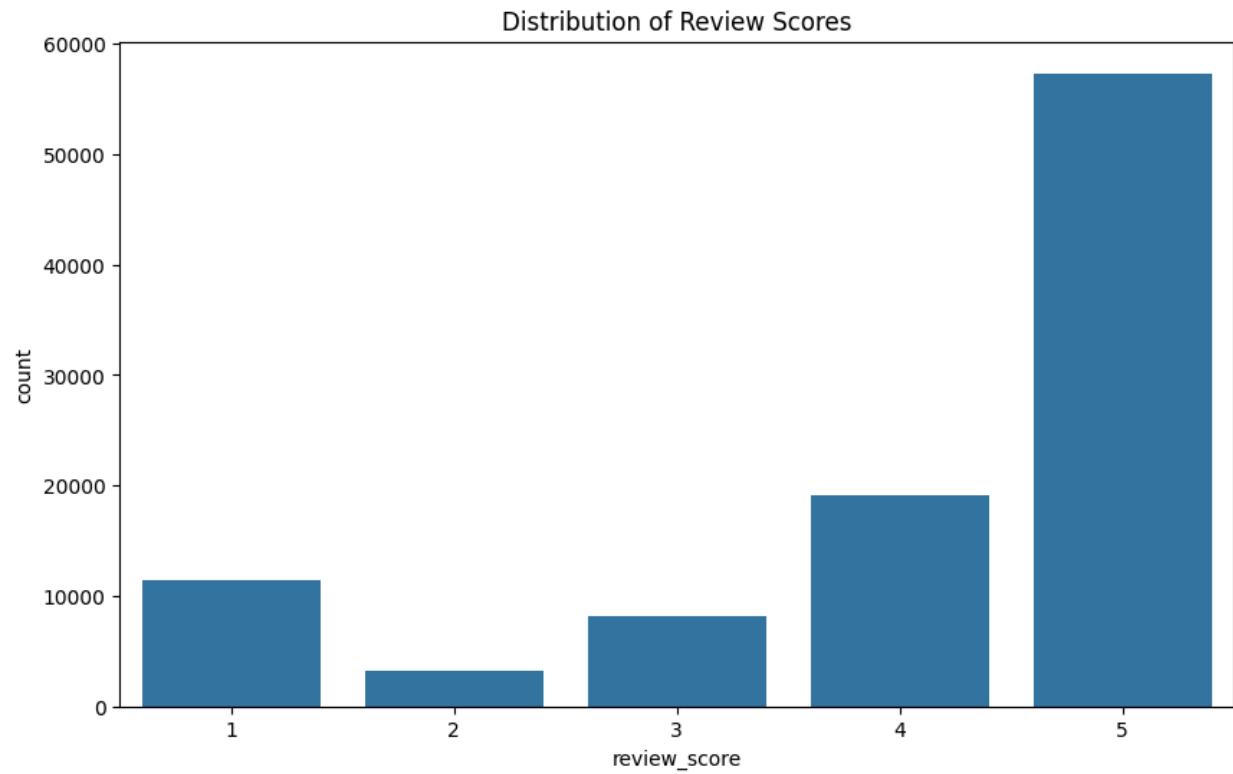
Box Plot of Payment Values



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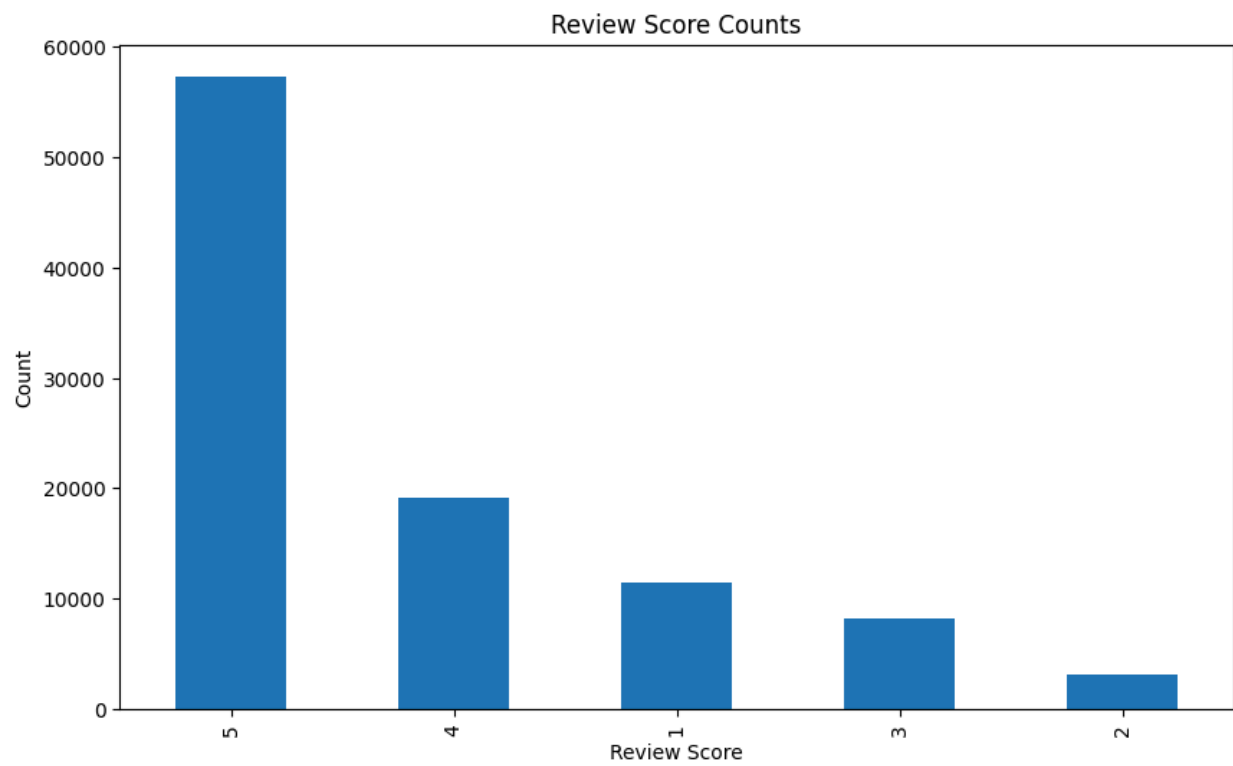
Order Reviews Table

Distribution of Review Scores



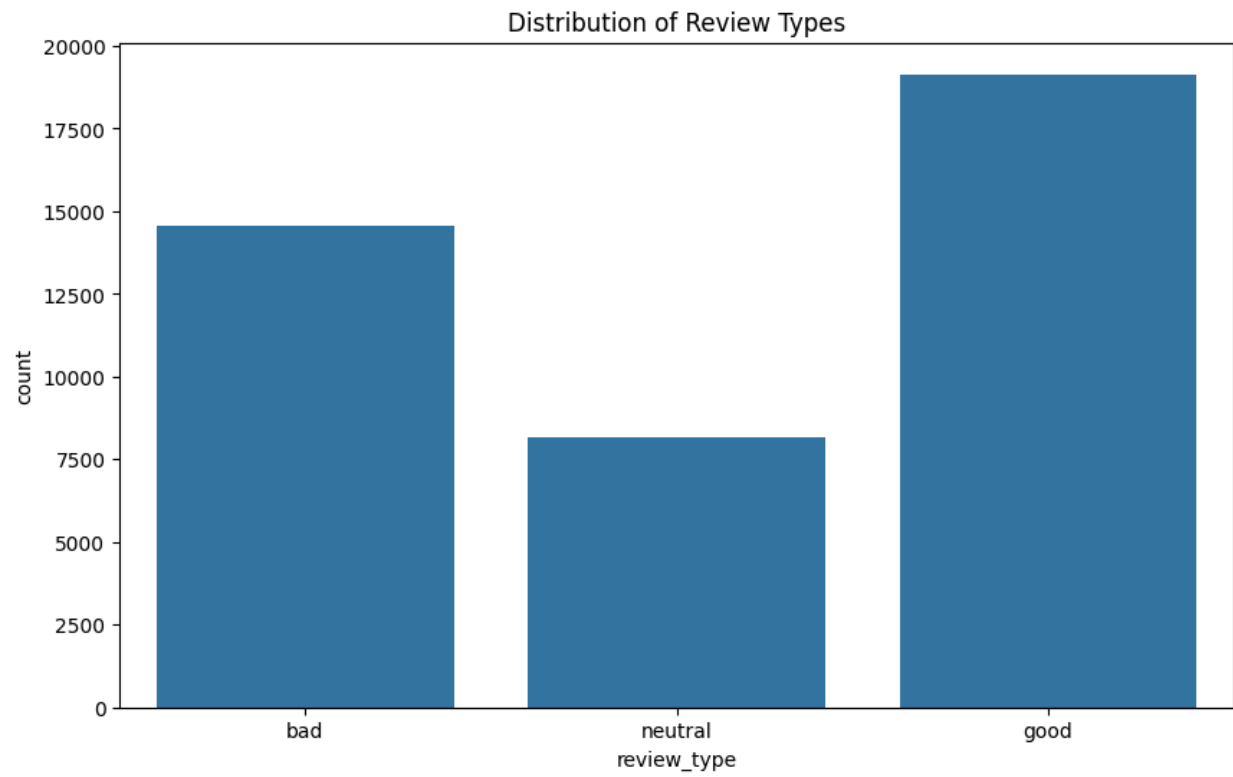
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Review Score Counts



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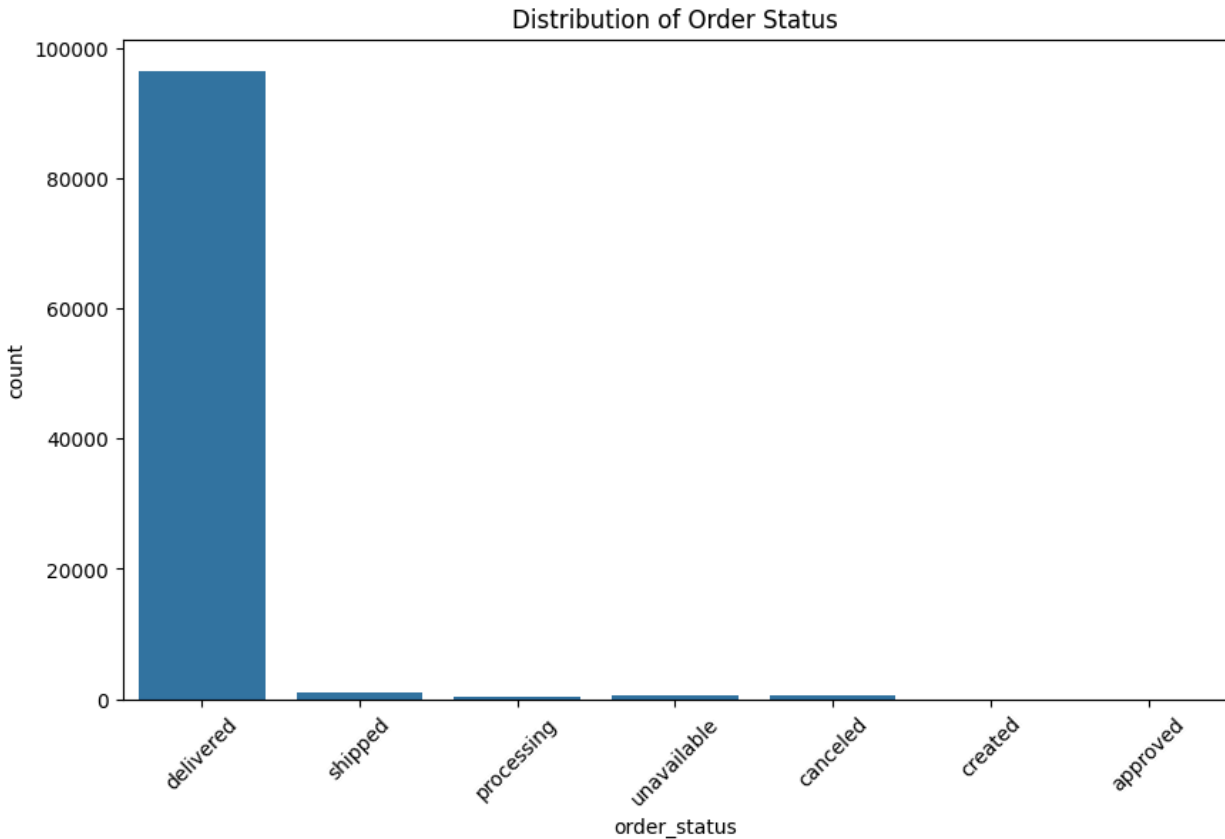
Distribution Review types



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Orders Table

Distribution of Order Status



Distribution of Order Delivery Time

- count 96461.000000
- mean 12.093582
- std 9.551576
- min 0.000000
- 25% 6.000000
- 50% 10.000000
- 75% 15.000000
- max 209.000000
- Name: delivery_time, dtype: float64

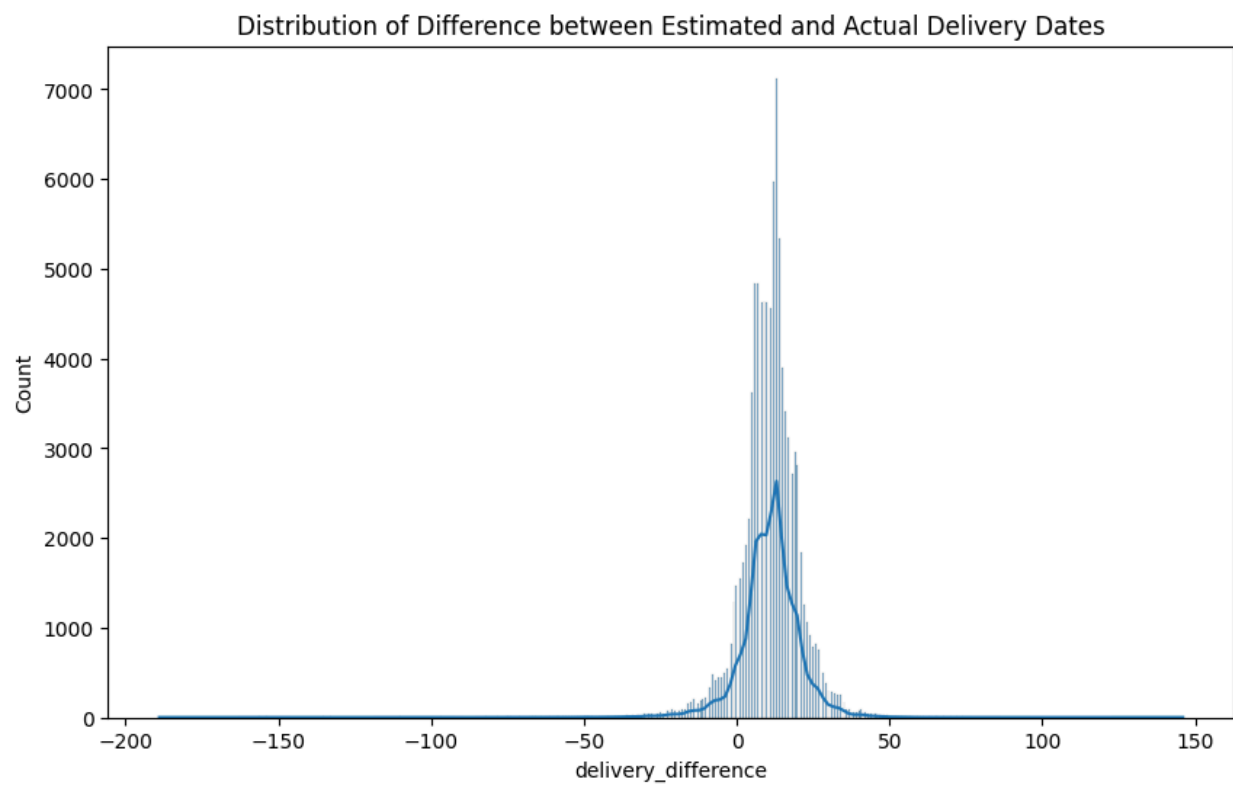
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Distribution of Difference between Estimated and Actual Delivery

- count 96461.000000
- mean 10.875577
- std 10.183476
- min -189.000000
- 25% 6.000000
- 50% 11.000000
- 75% 16.000000
- max 146.000000

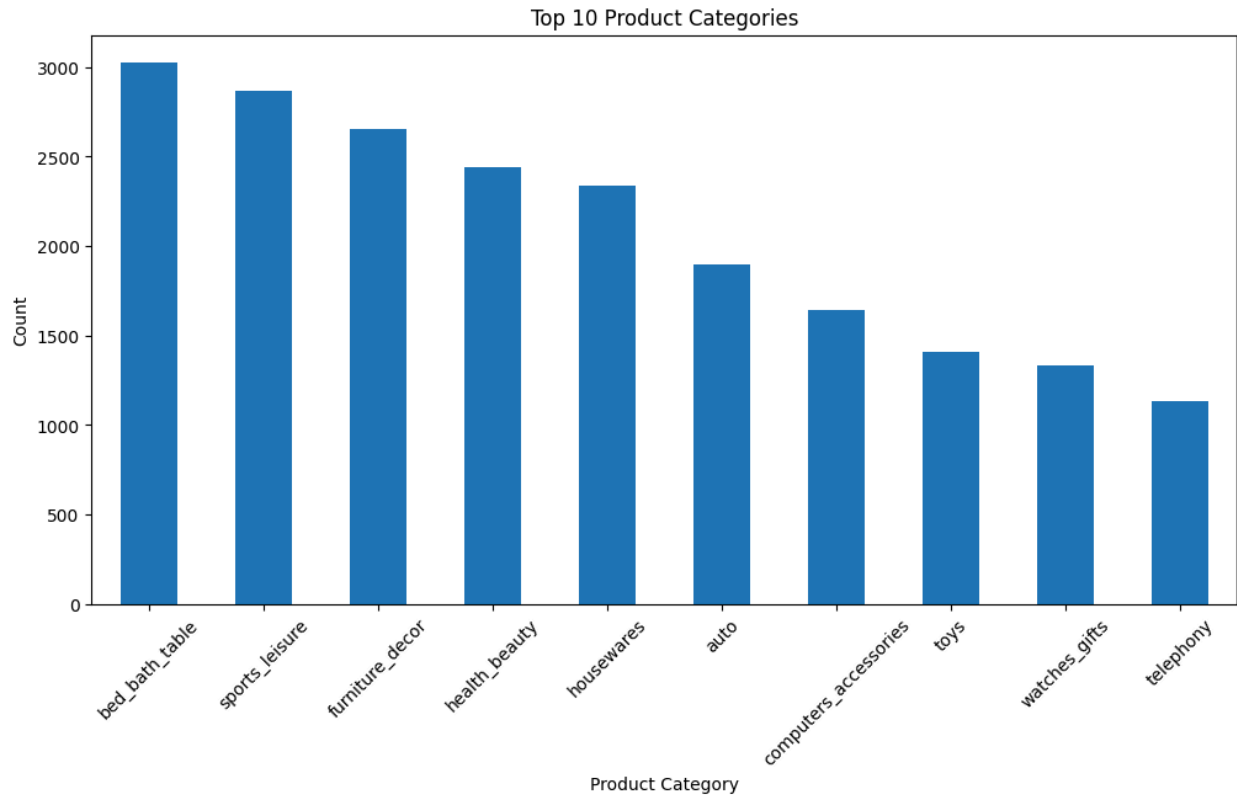
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Sravani Garapati (11936780)

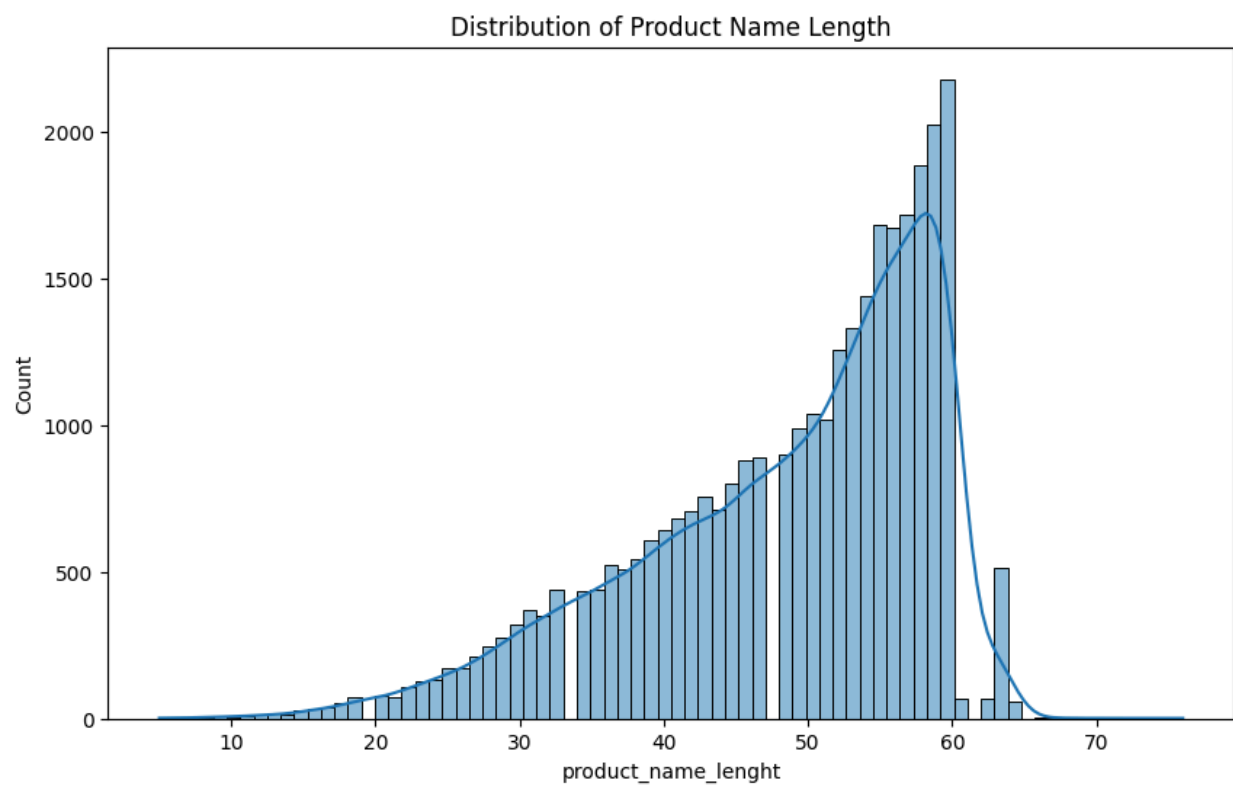
Products Table

Top 10 Product Categories



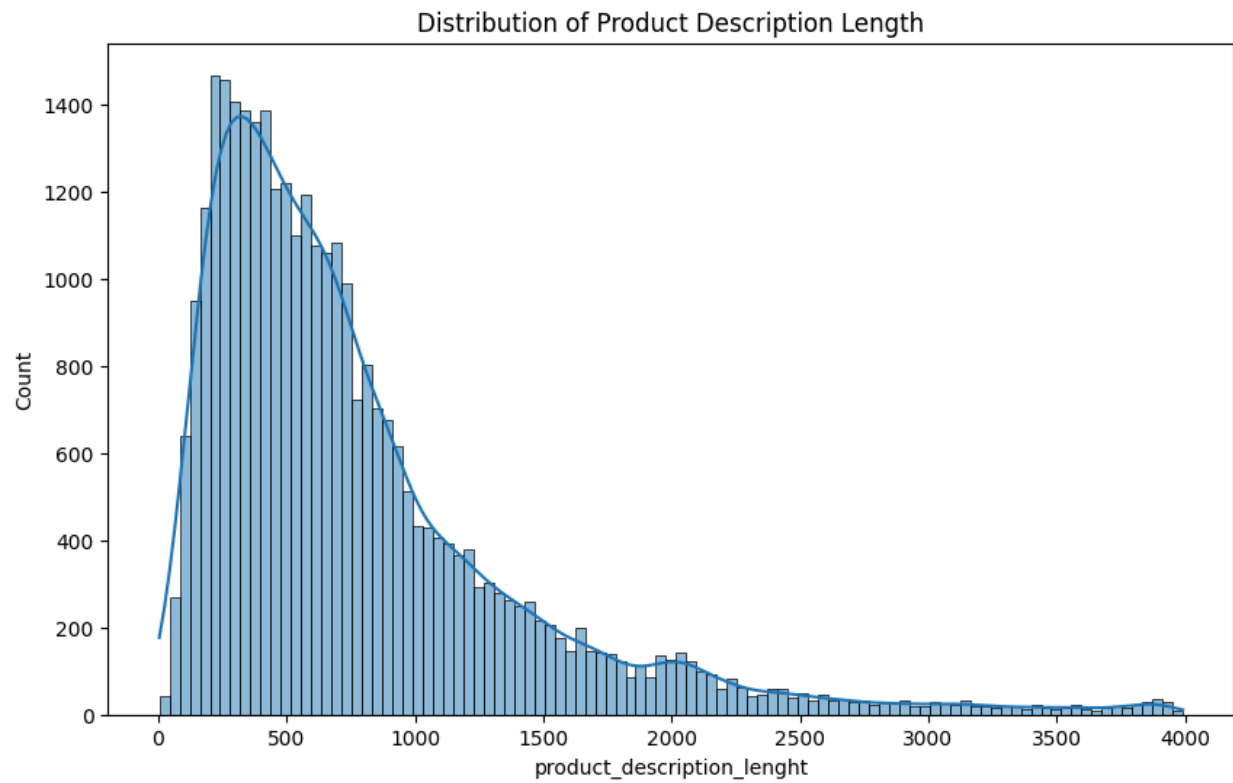
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Distribution of Product Name Length



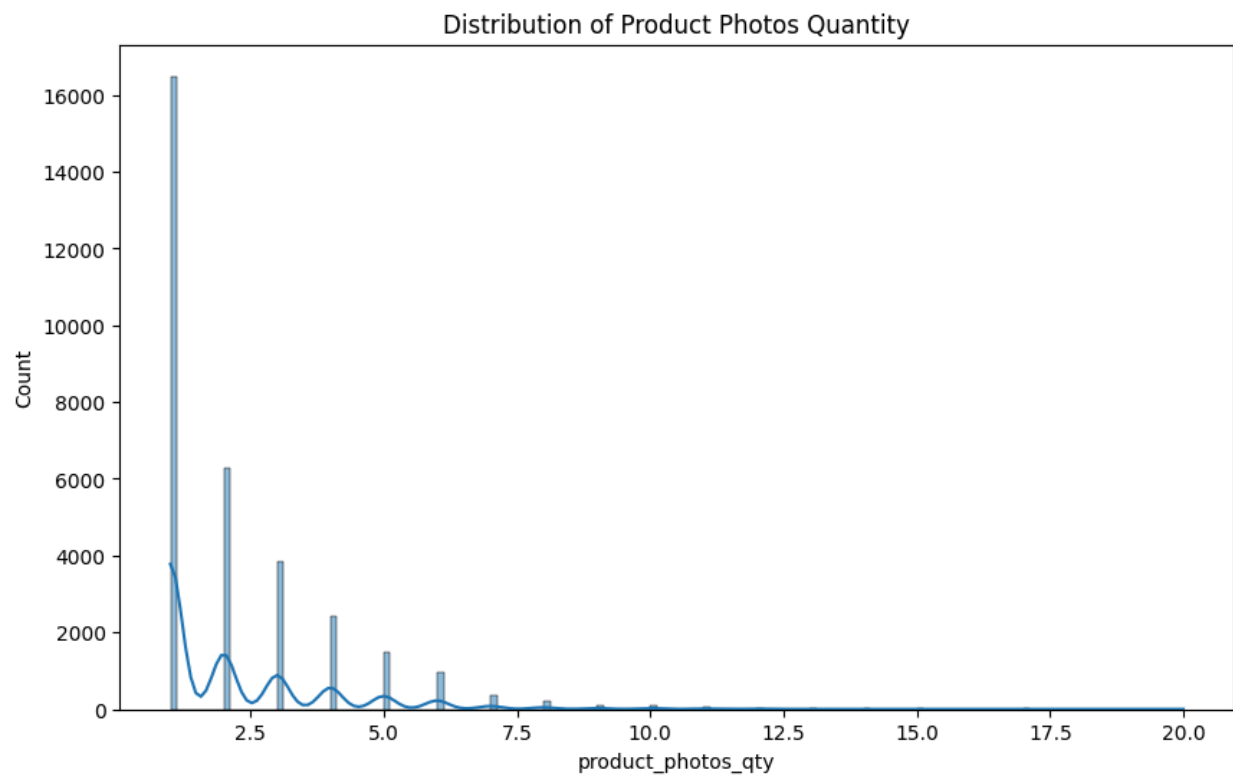
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Distribution of Product Description Length



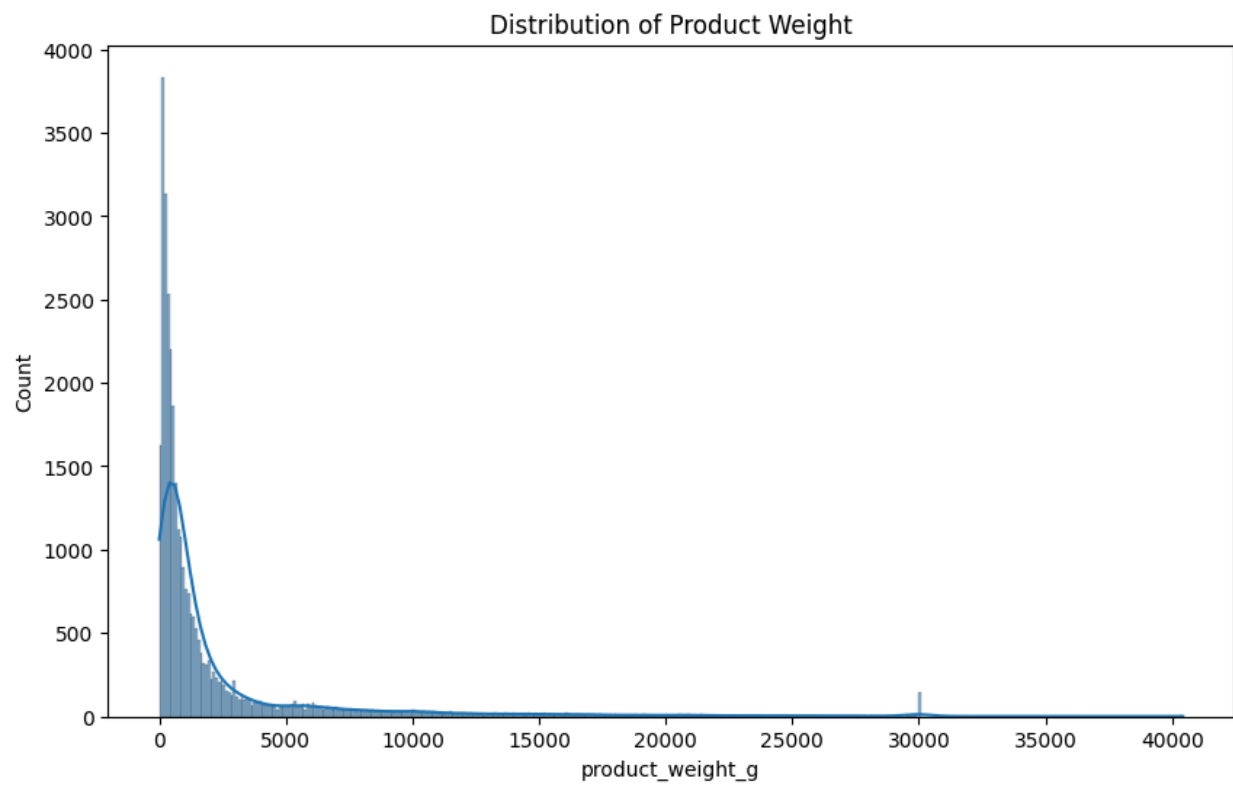
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Distribution of Product Photos Quantity



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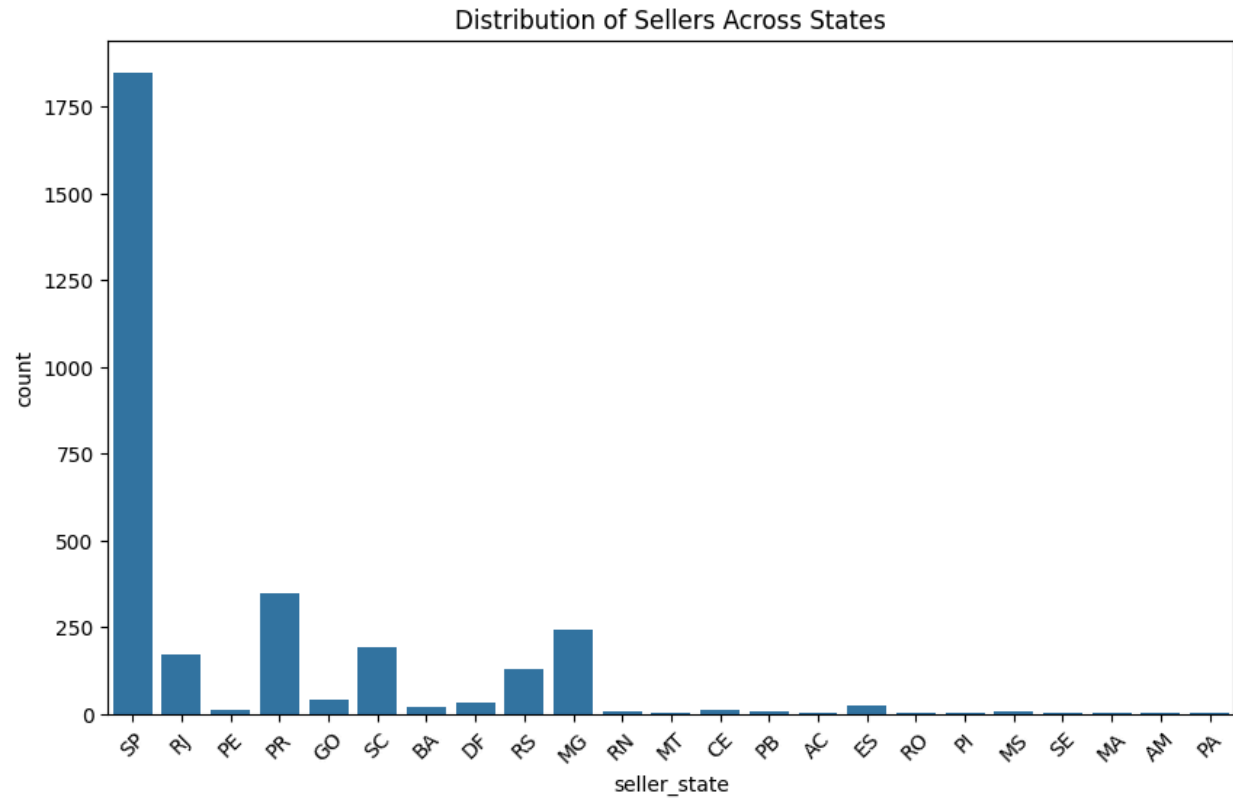
Distribution of Product Weight



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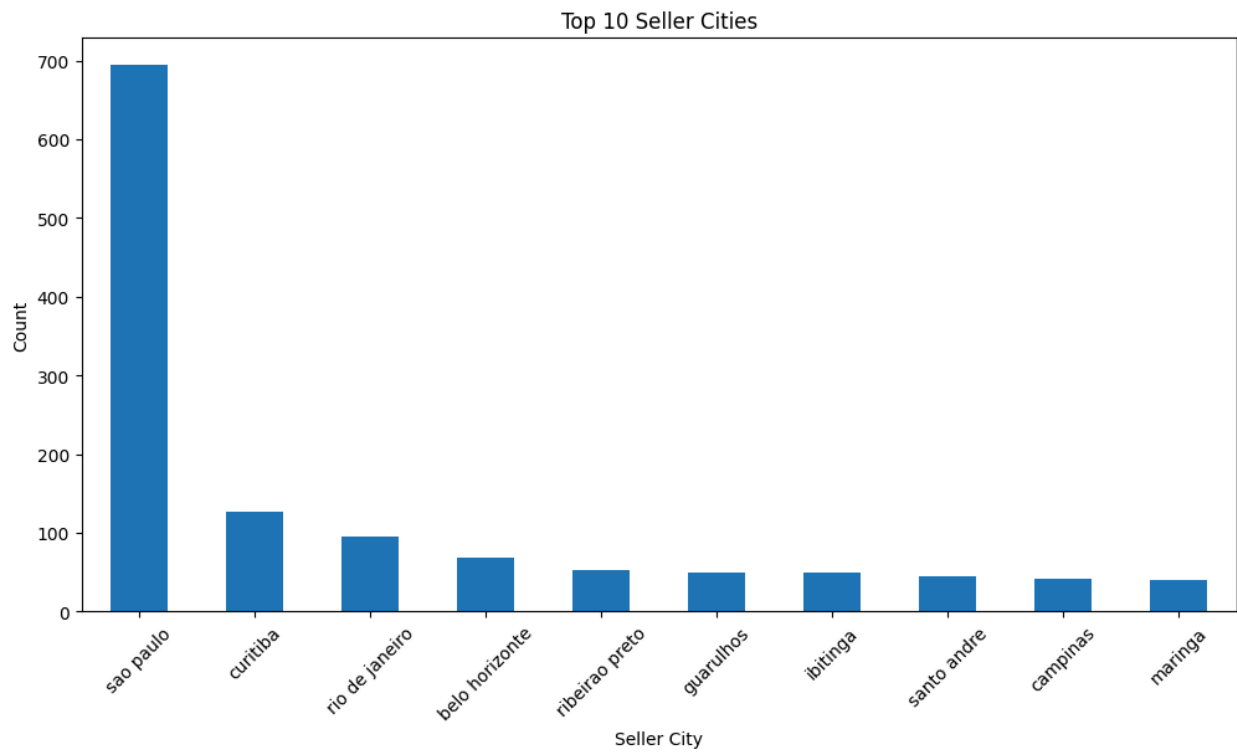
Sellers Table

Distribution of Seller Across the States



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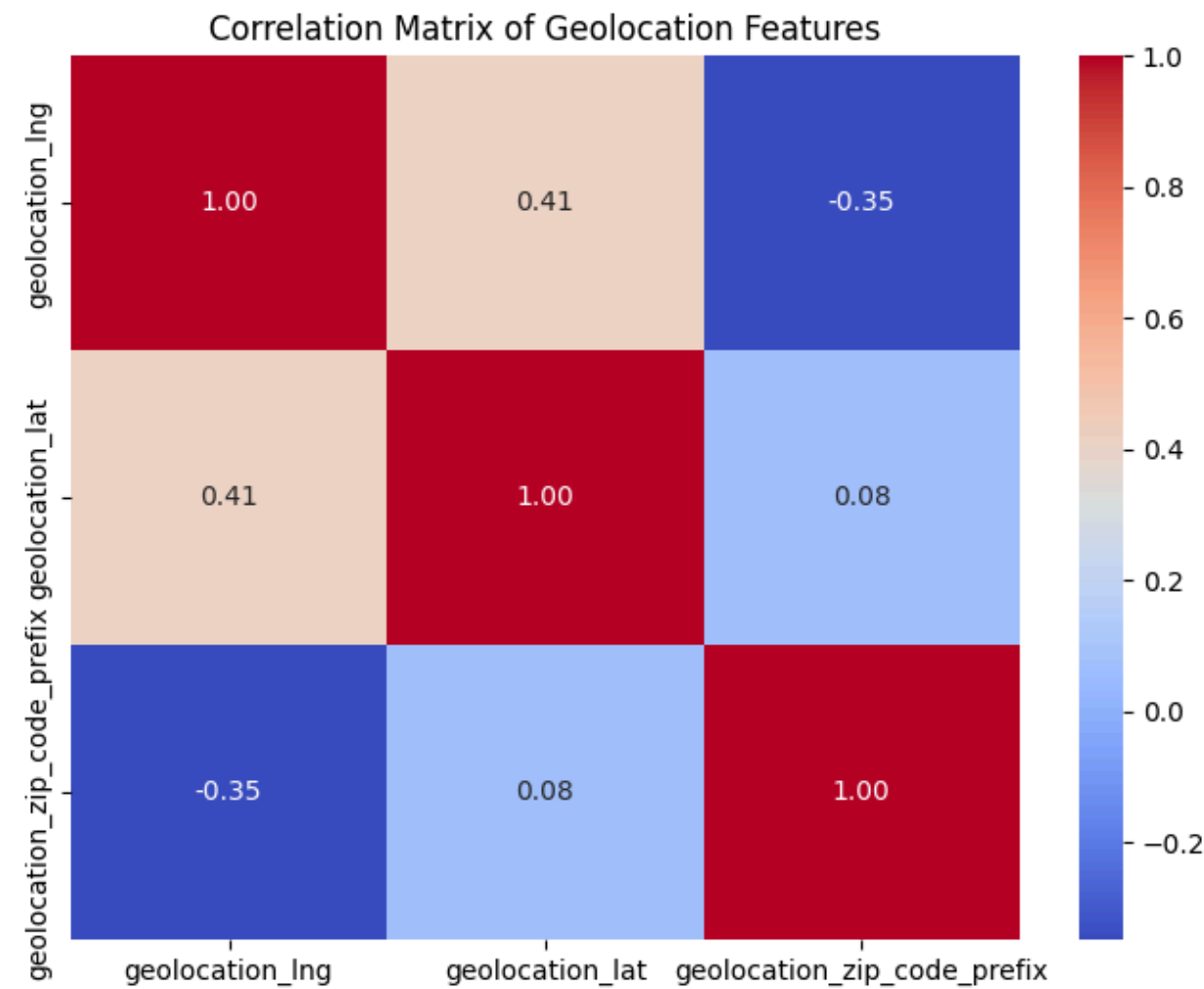
Top 10 Seller Cities



Column Correlations

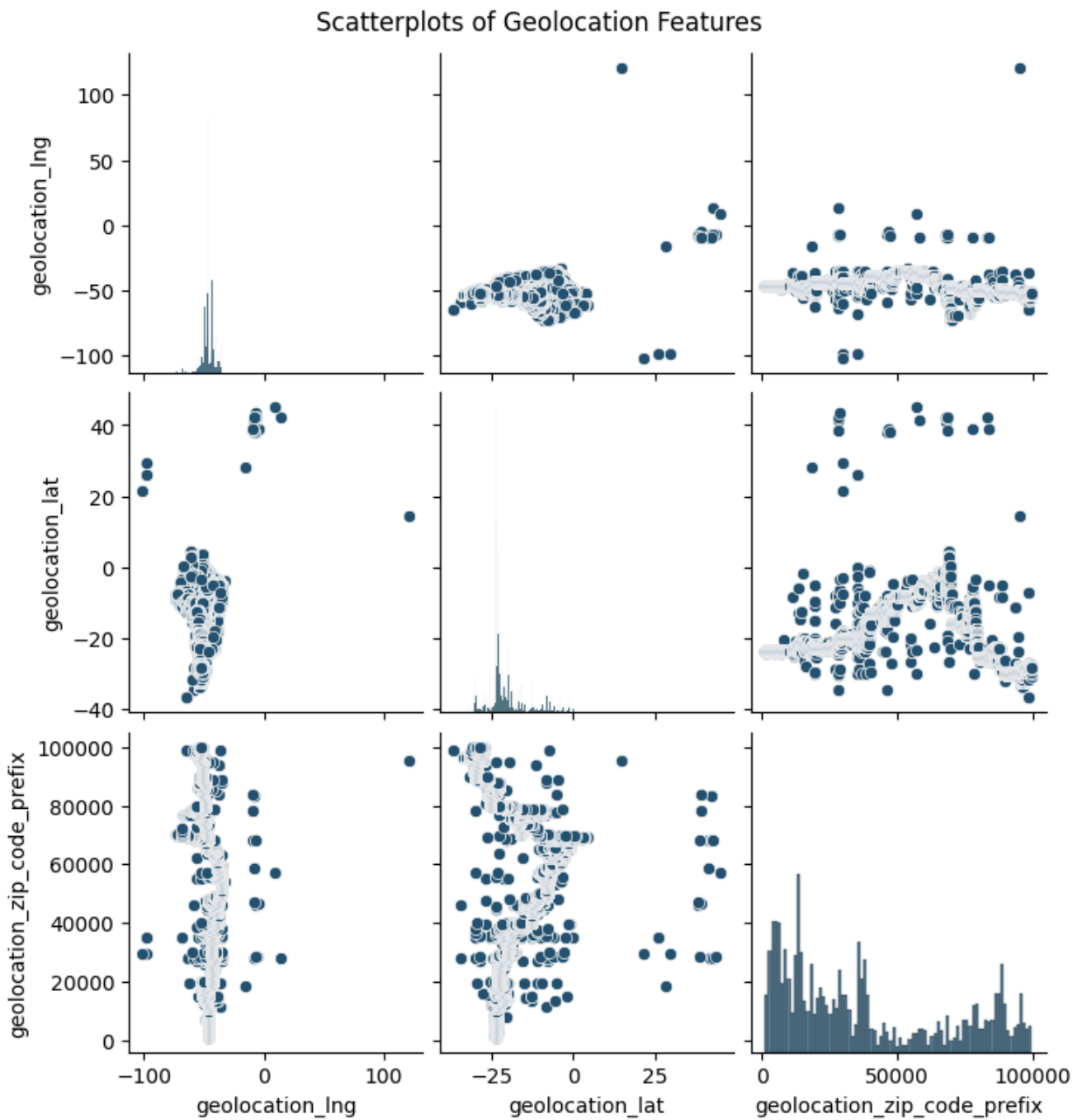
Geolocations Table

Correlation Matrix of Geolocation Features



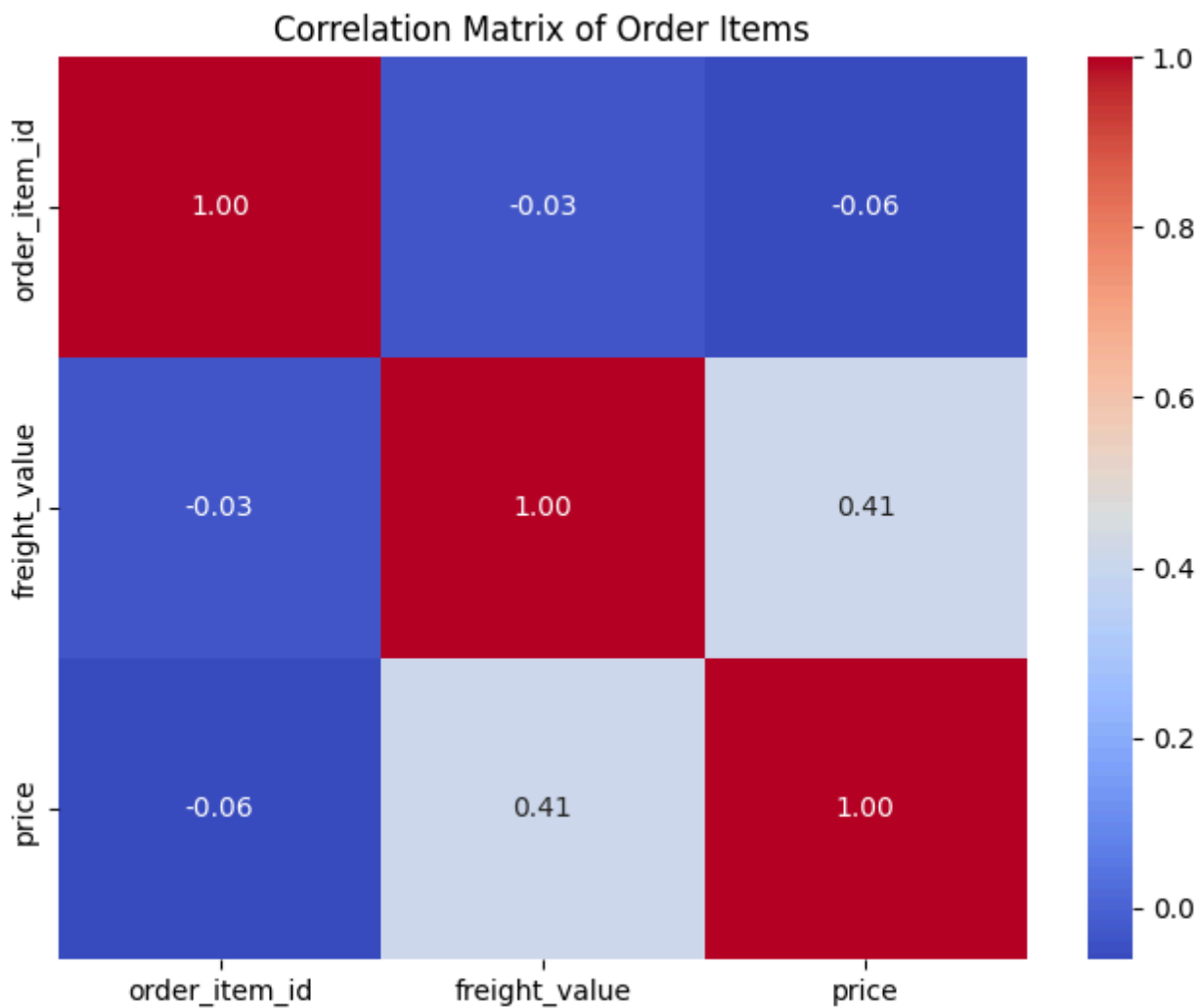
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Scatterplots of Geolocation Features



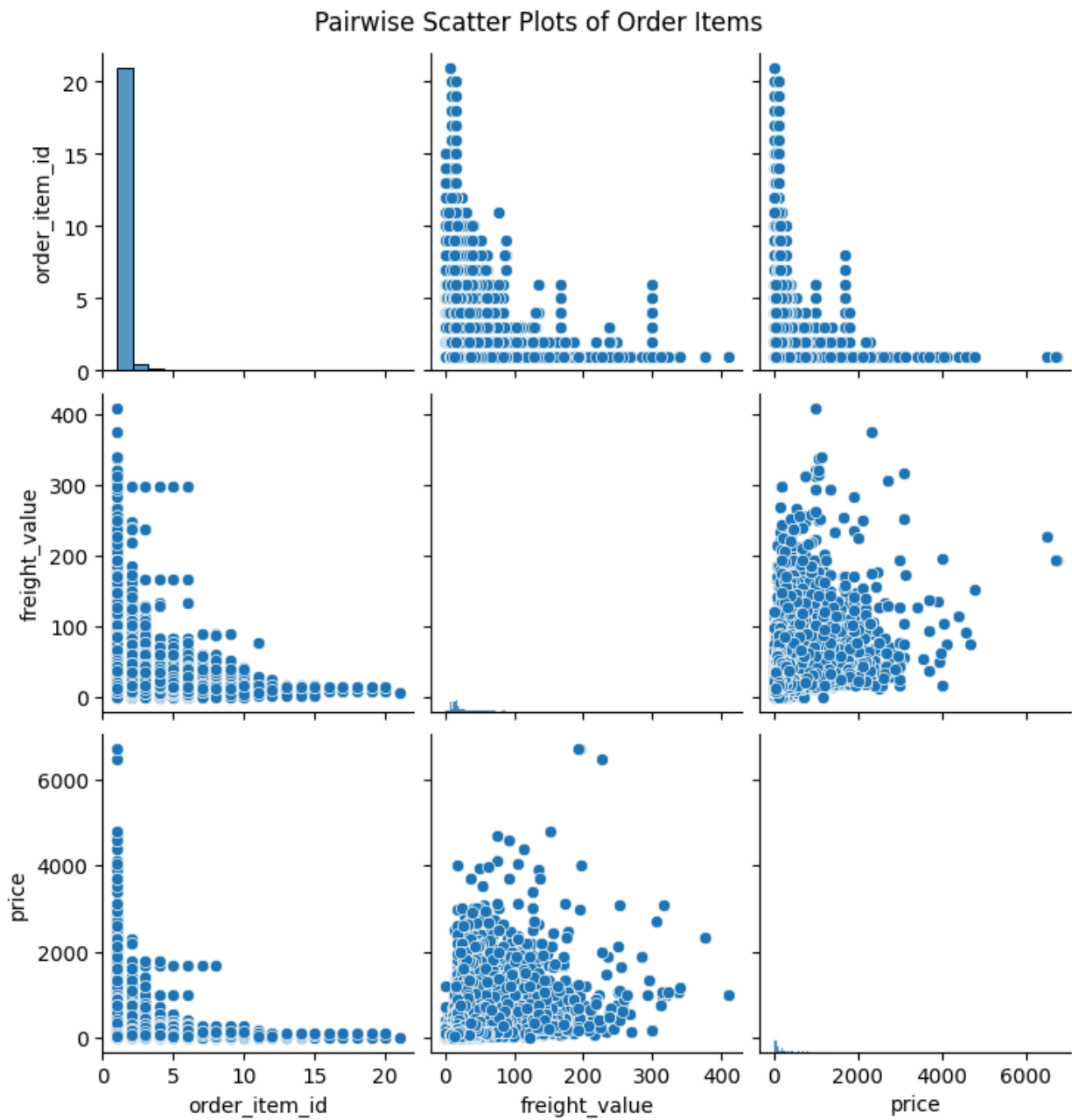
Order Items Table

Correlation Matrix of Order Items



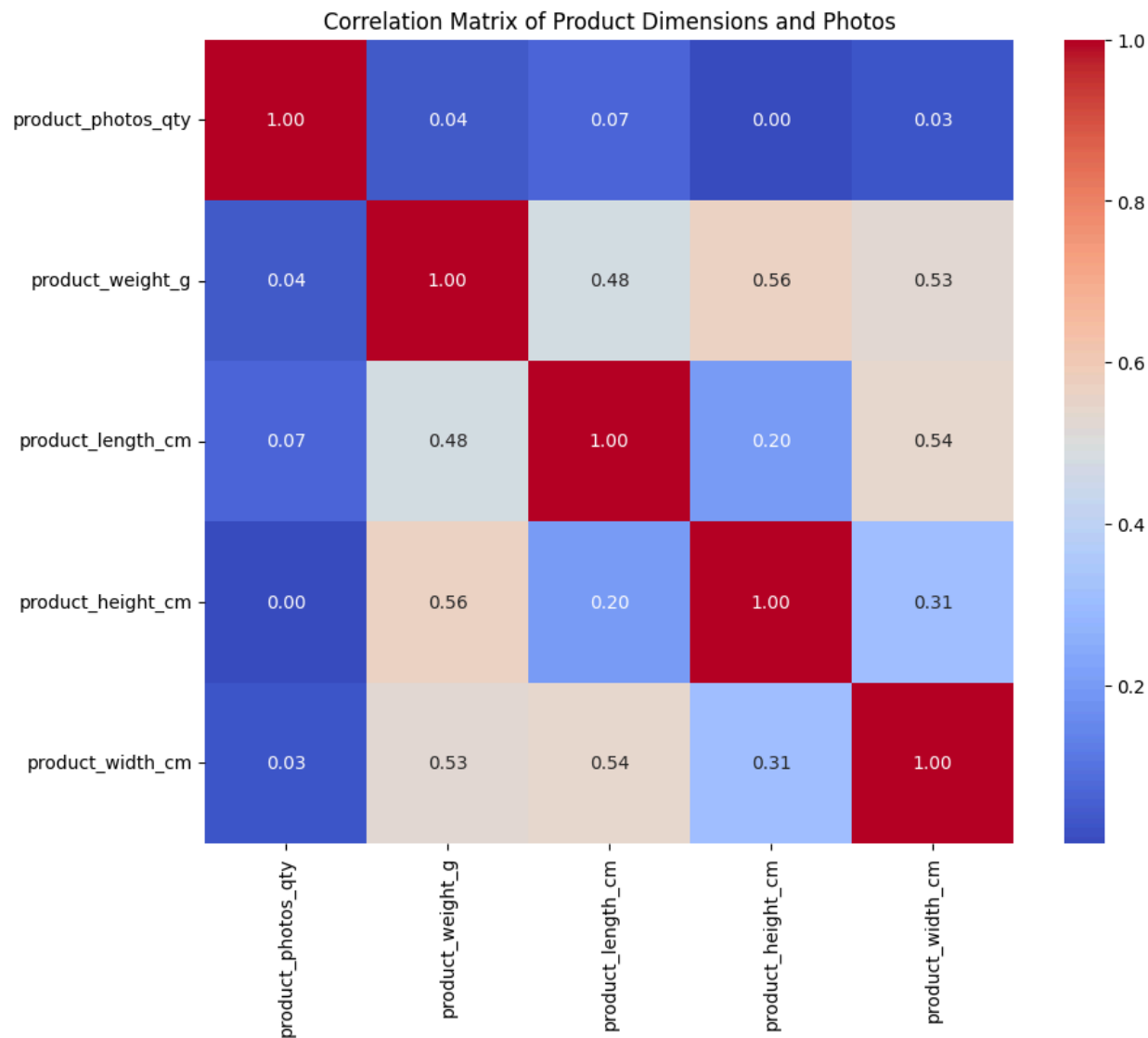
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Pairwise Scatter Plots of Order Items



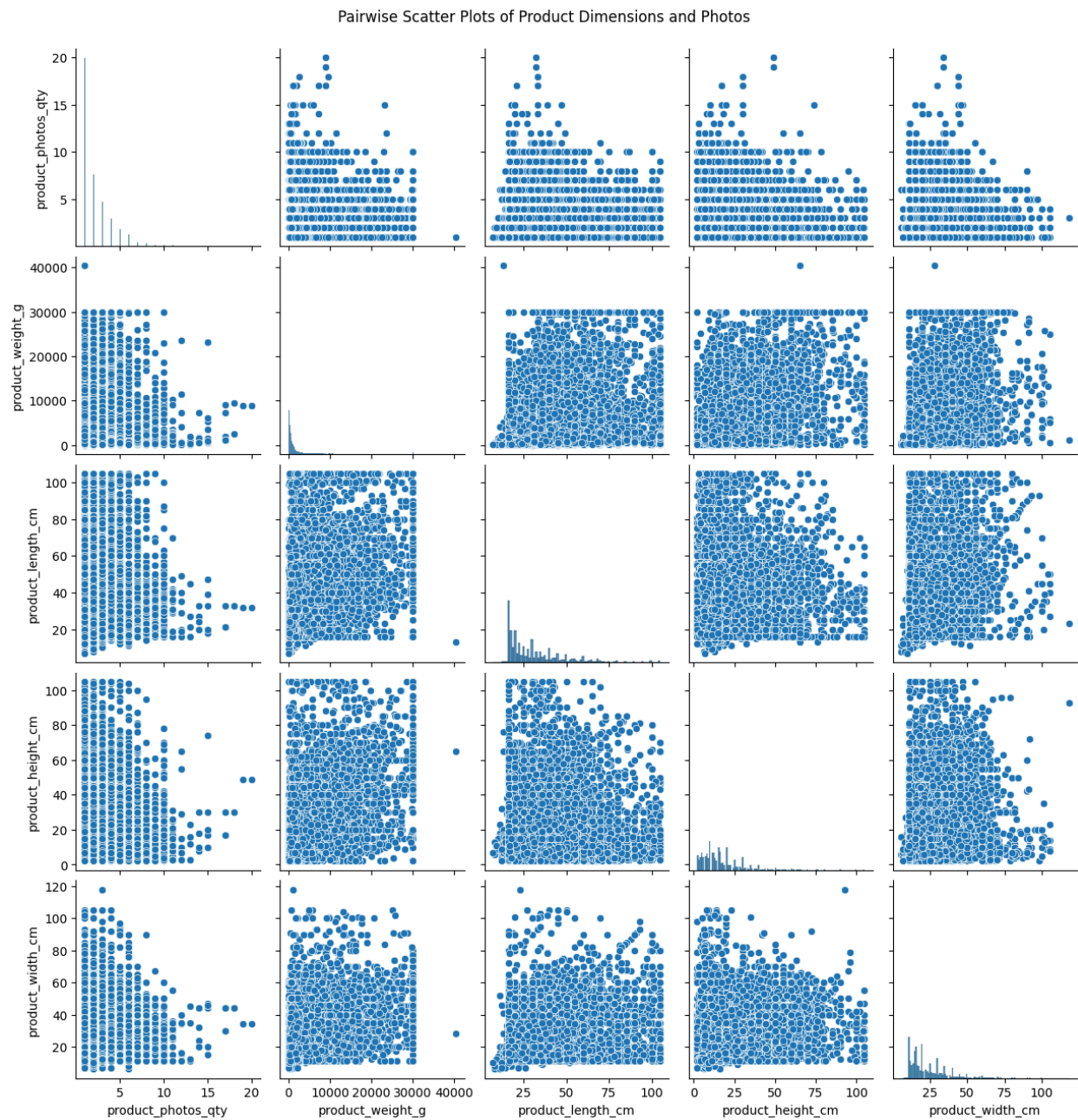
Products Table

Correlation Matrix of Product Dimensions and Photos



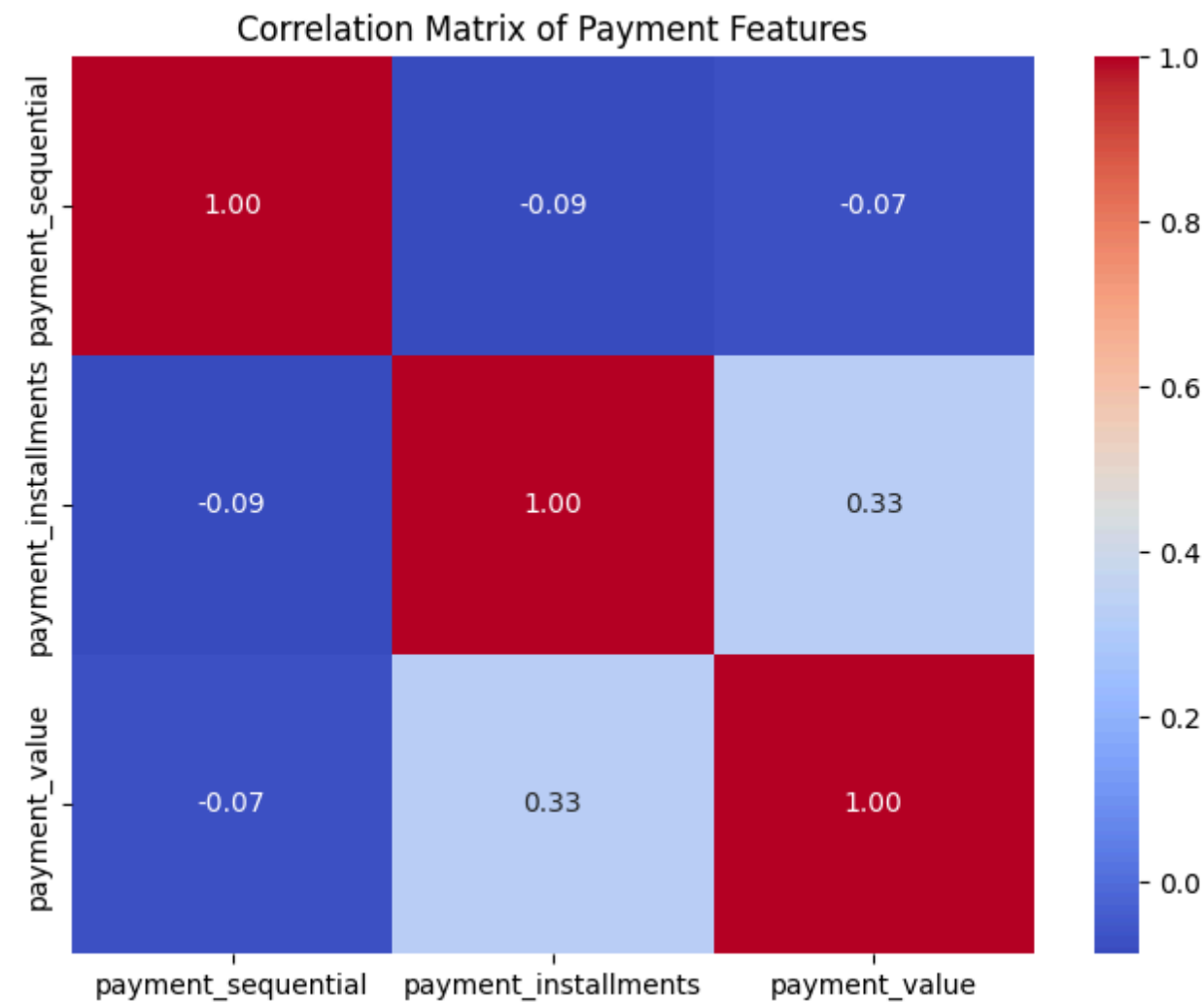
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Pairwise Scatter Plots of Product Dimensions and Photos



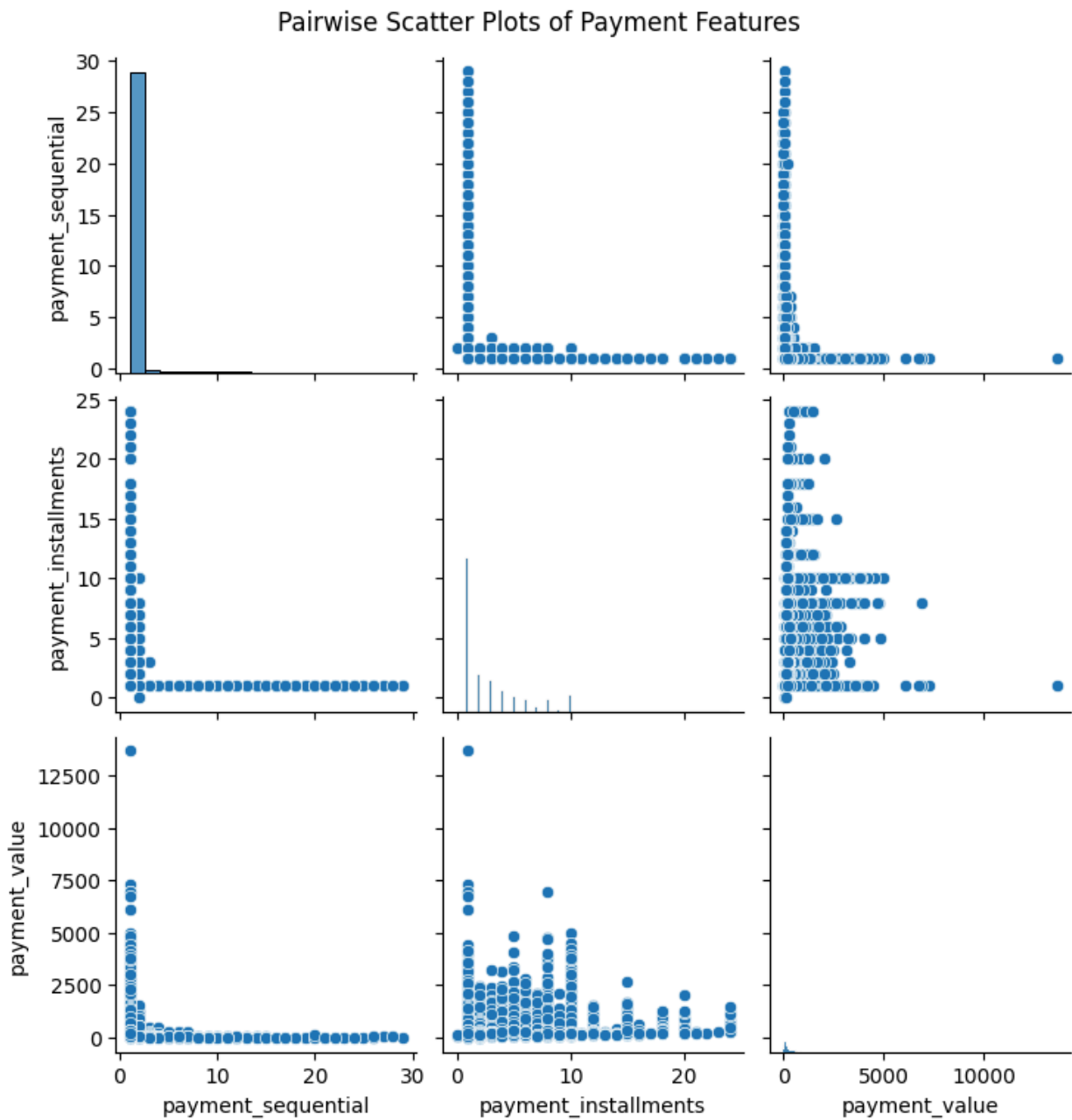
Payments Table

Correlation Matrix of Payment Features



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Pairwise Scatter Plots of Payment Features



Joining All Tables

RangeIndex: 106556 entries, 0 to 106555

Data columns (total 42 columns):

#	Column	Non-Null Count	Dtype
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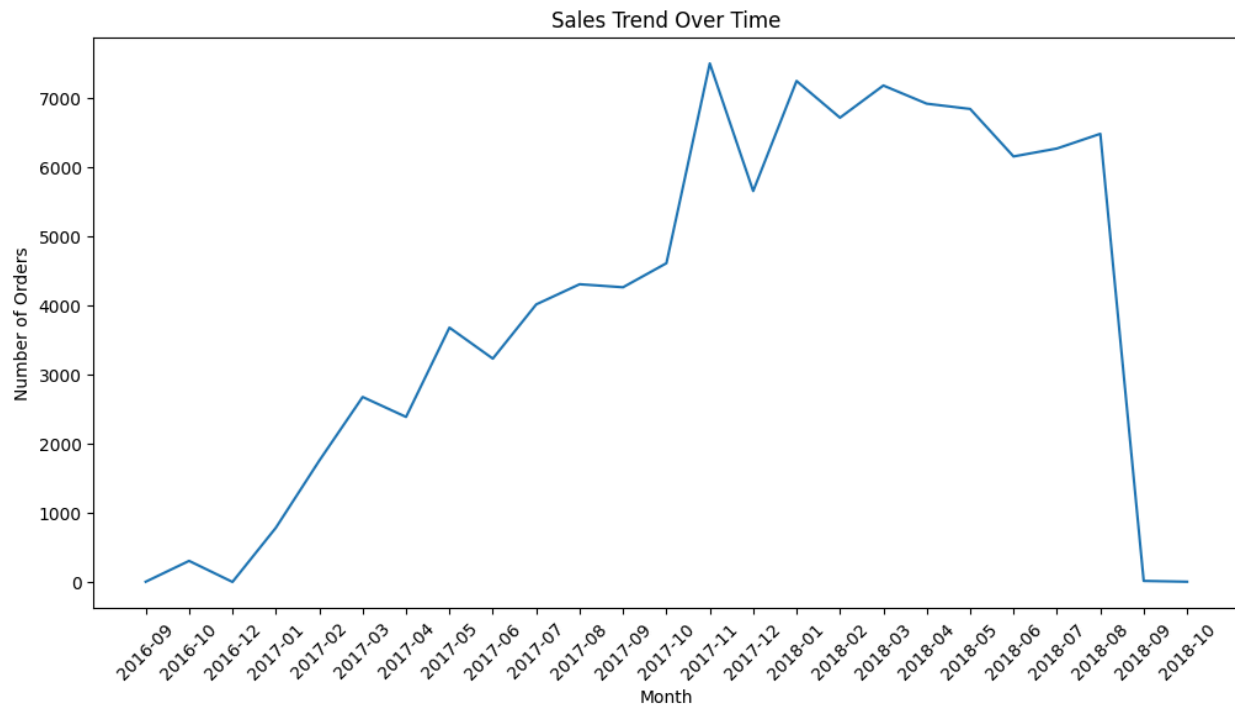
0	order_id	106556 non-null	object
1	customer_id	106556 non-null	object
2	order_status	106556 non-null	object
3	order_purchase_timestamp	106556 non-null	datetime64[ns]
4	order_approved_at	106556 non-null	datetime64[ns]
5	order_delivered_carrier_date	105791 non-null	datetime64[ns]
6	order_delivered_customer_date	104683 non-null	datetime64[ns]
7	order_estimated_delivery_date	106556 non-null	datetime64[ns]
8	delivery_time	104683 non-null	float64
9	delivery_difference	104683 non-null	float64
10	review_id	106556 non-null	object
11	review_score	106556 non-null	int64
12	review_comment_title	12675 non-null	object
13	review_comment_message	45057 non-null	object
14	review_creation_date	106556 non-null	object
15	review_answer_timestamp	106556 non-null	object
16	review_type	46167 non-null	category
17	payment_sequential	106556 non-null	int64
18	payment_type	106556 non-null	object
19	payment_installments	106556 non-null	int64
20	payment_value	106556 non-null	float64
21	customer_unique_id	106556 non-null	object
22	customer_zip_code_prefix	106556 non-null	int64
23	customer_city	106556 non-null	object
24	customer_state	106556 non-null	object
25	order_item_id	106556 non-null	int64
26	product_id	106556 non-null	object
27	seller_id	106556 non-null	object
28	shipping_limit_date	106556 non-null	object
29	price	106556 non-null	float64
30	freight_value	106556 non-null	float64
31	product_category_name	106556 non-null	object
32	product_name_lenght	106556 non-null	float64
33	product_description_lenght	106556 non-null	float64
34	product_photos_qty	106556 non-null	float64
35	product_weight_g	106556 non-null	float64
36	product_length_cm	106556 non-null	float64
37	product_height_cm	106556 non-null	float64
38	product_width_cm	106556 non-null	float64

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```
39 seller_zip_code_prefix    106556 non-null int64
40 seller_city                106556 non-null object
41 seller_state               106556 non-null object
dtypes: category(1), datetime64[ns](5), float64(12), int64(6), object(18)
memory usage: 33.4+ MB
```

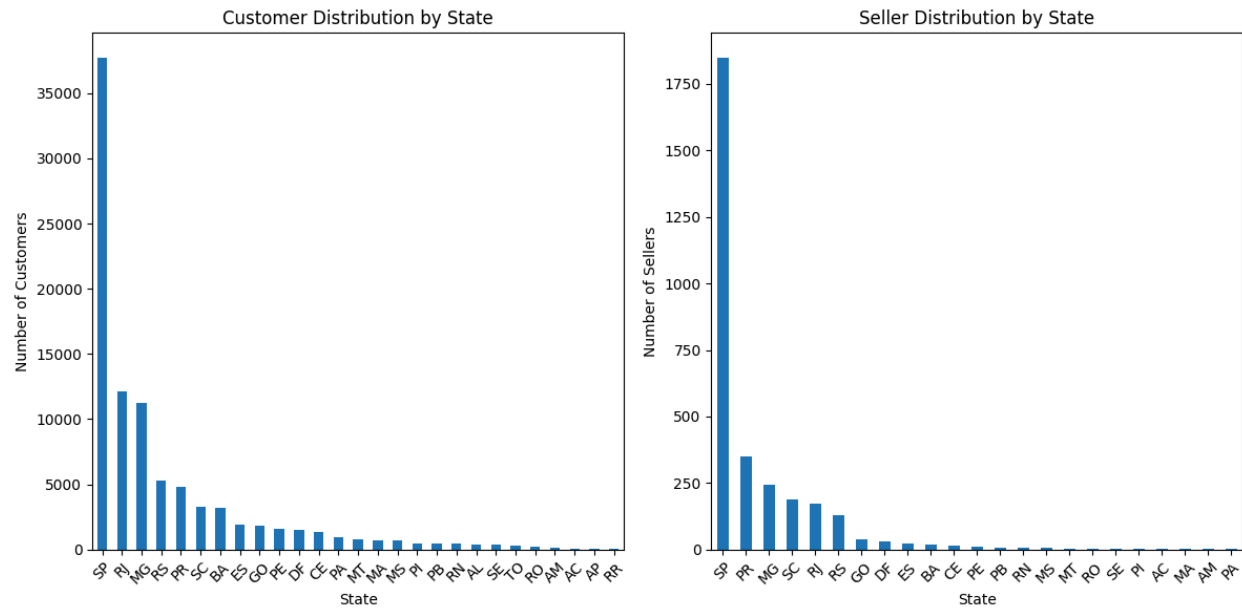
Exploratory Data Analysis

Sales Trend Over Time



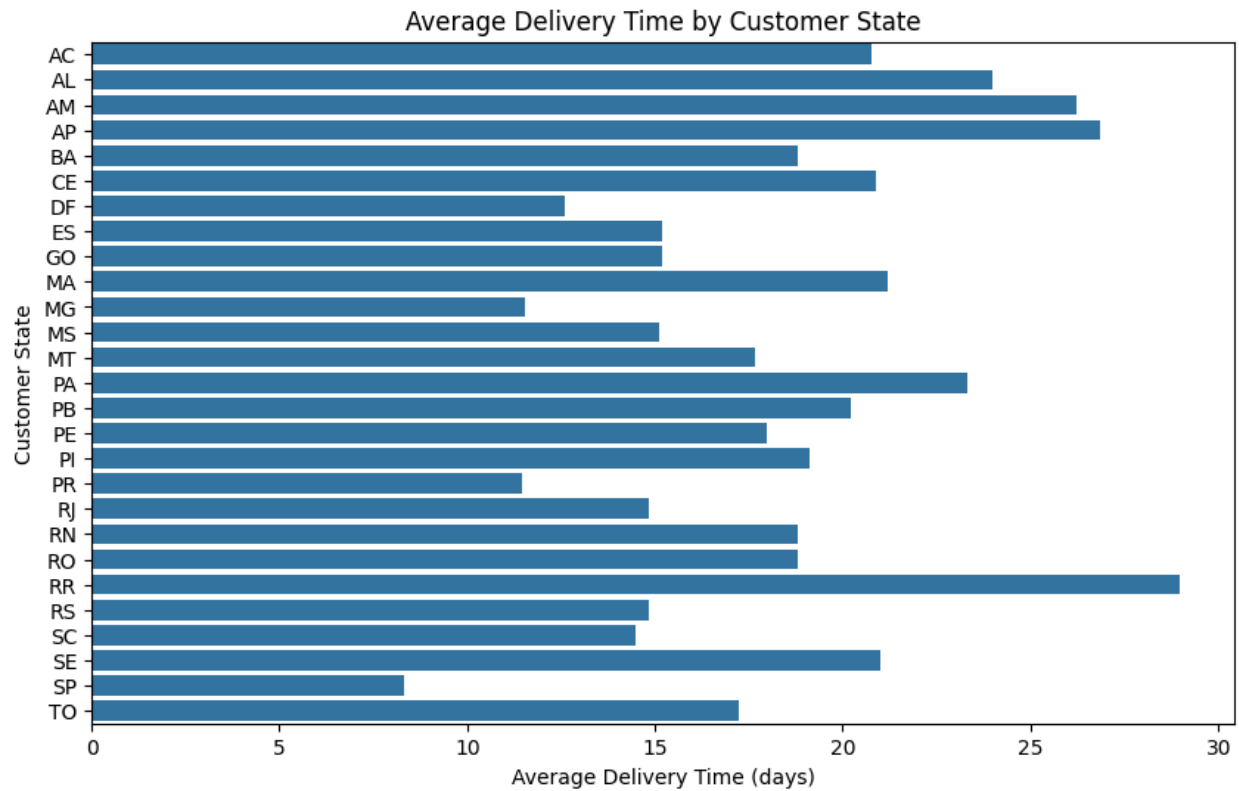
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Customer and Seller Distribution



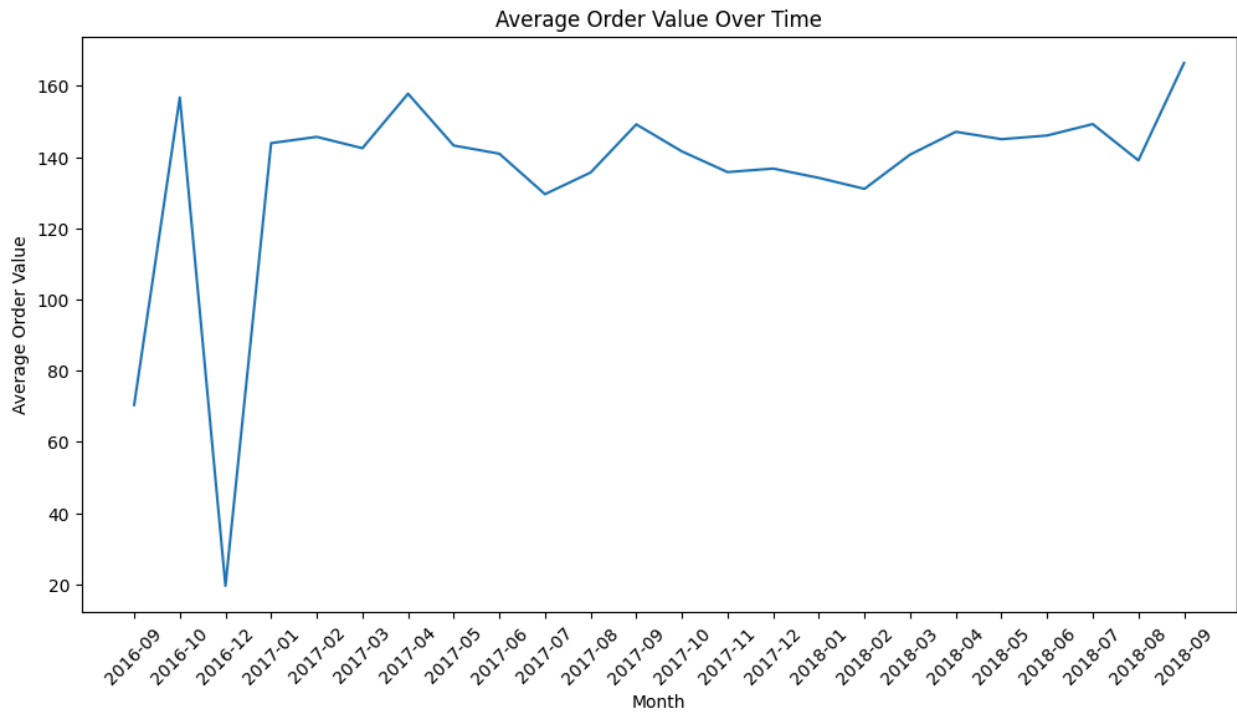
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Sravani Garapati (11936780)

Average Delivery Time by Customer State



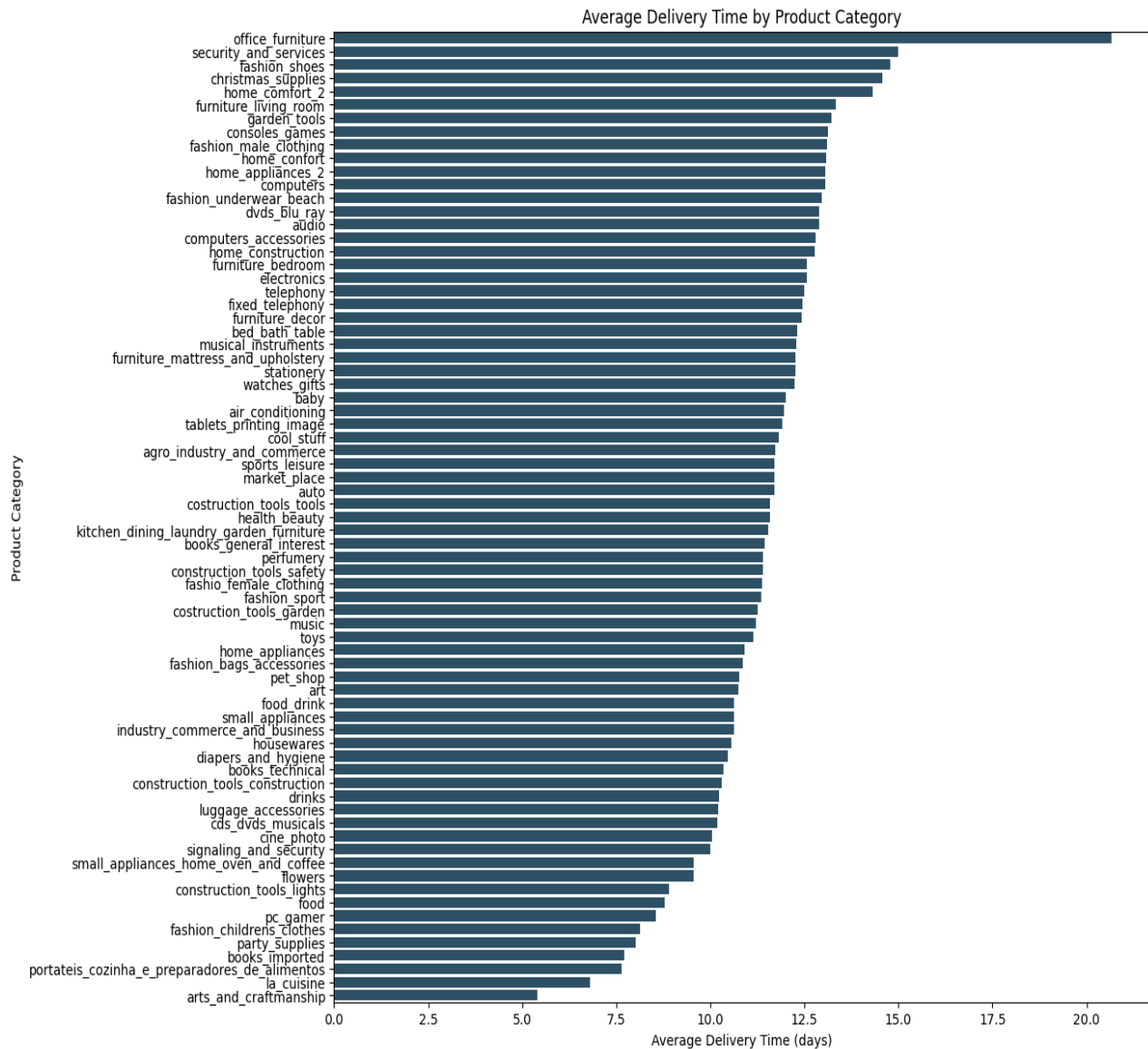
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Average Order Value Over Time



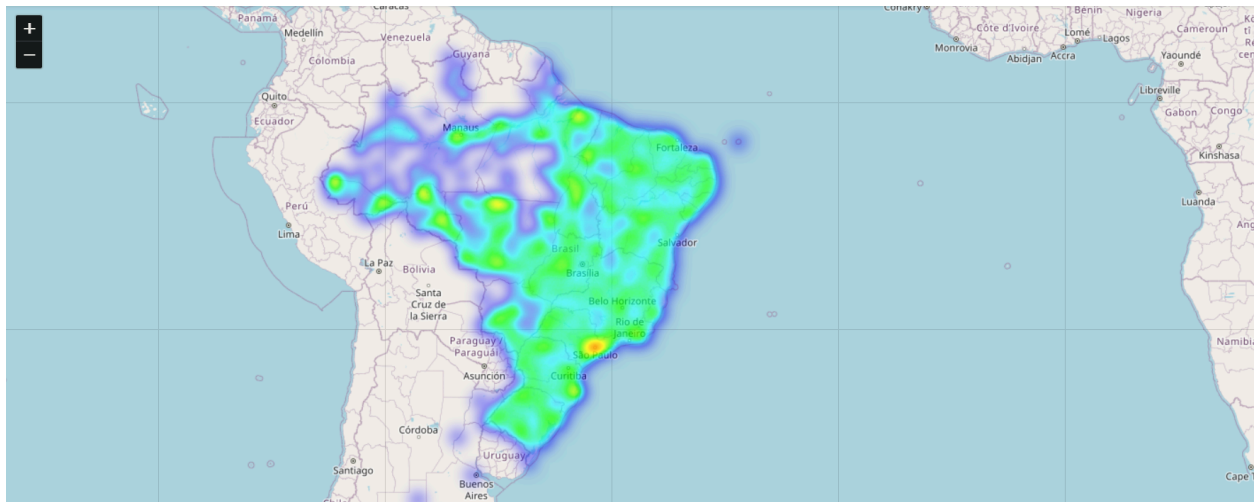
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Average Delivery Time by Category

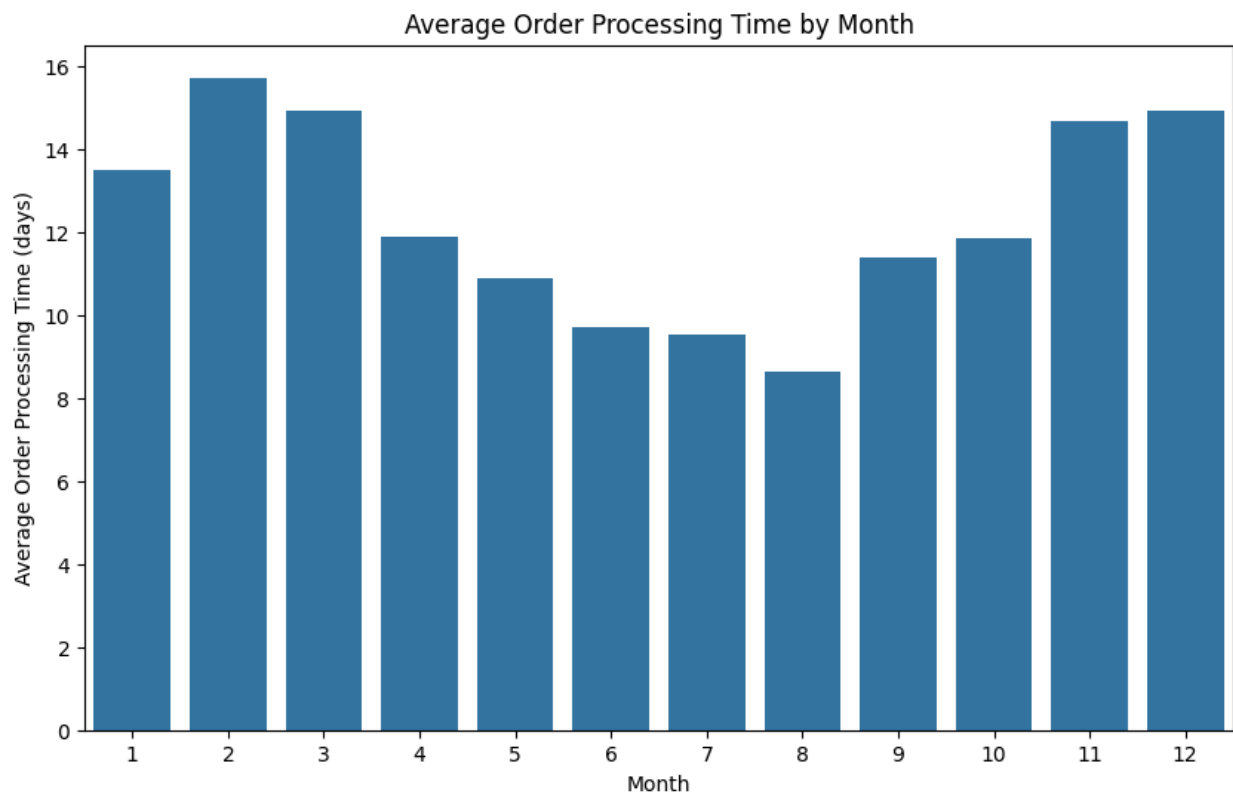


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Geolocation Map

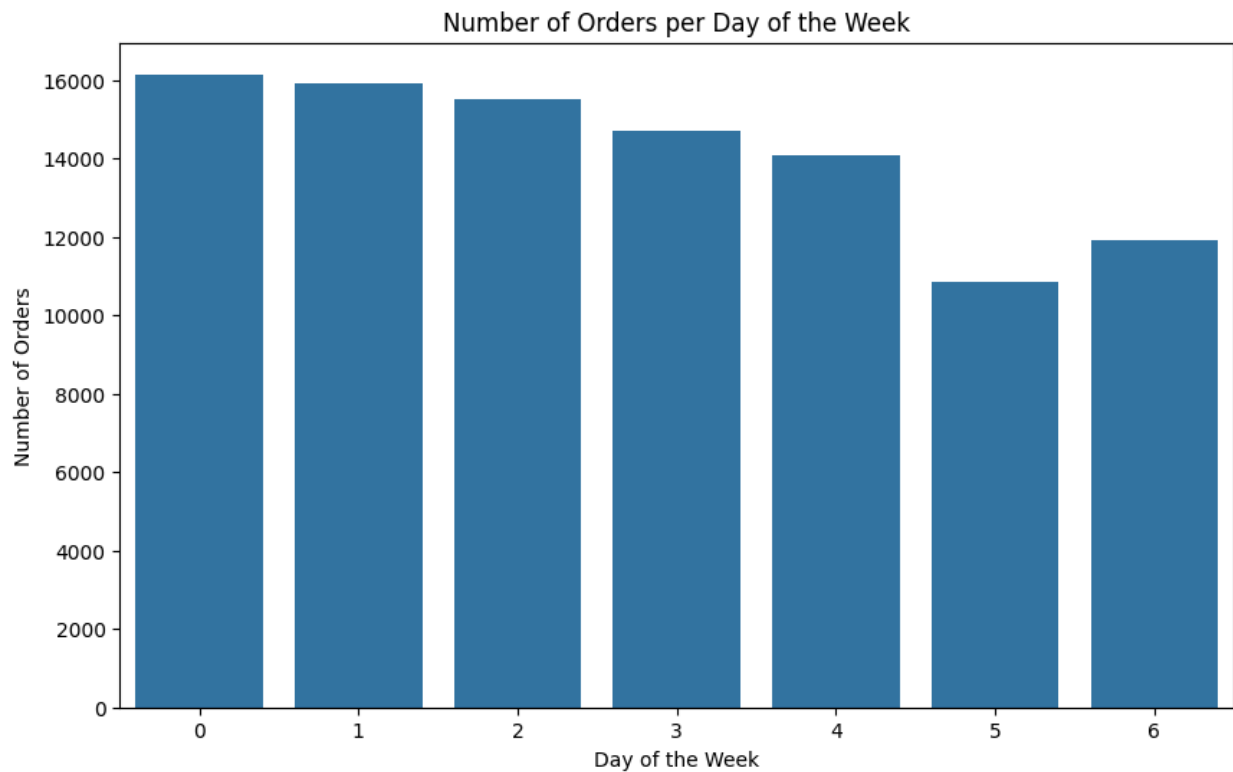


Average Processing Time by Month



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Number of Orders per Day of the Week

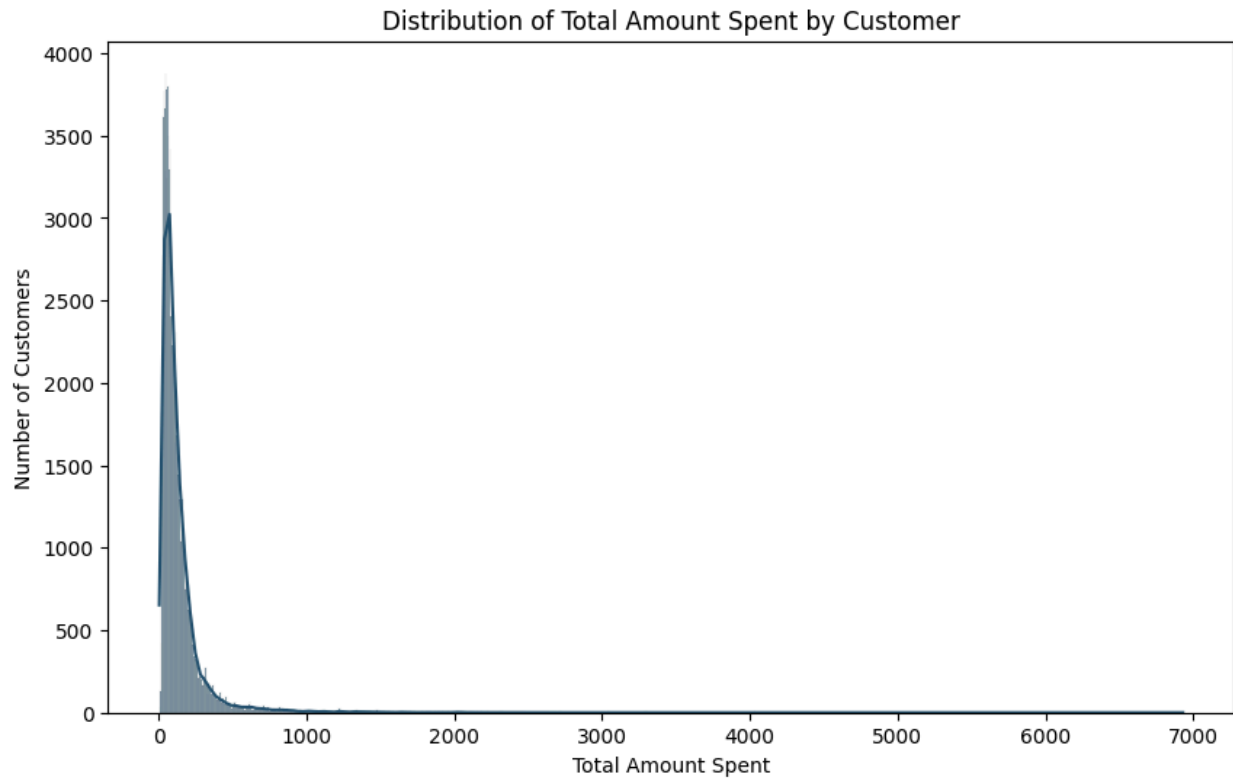


Word Cloud of Review Comments



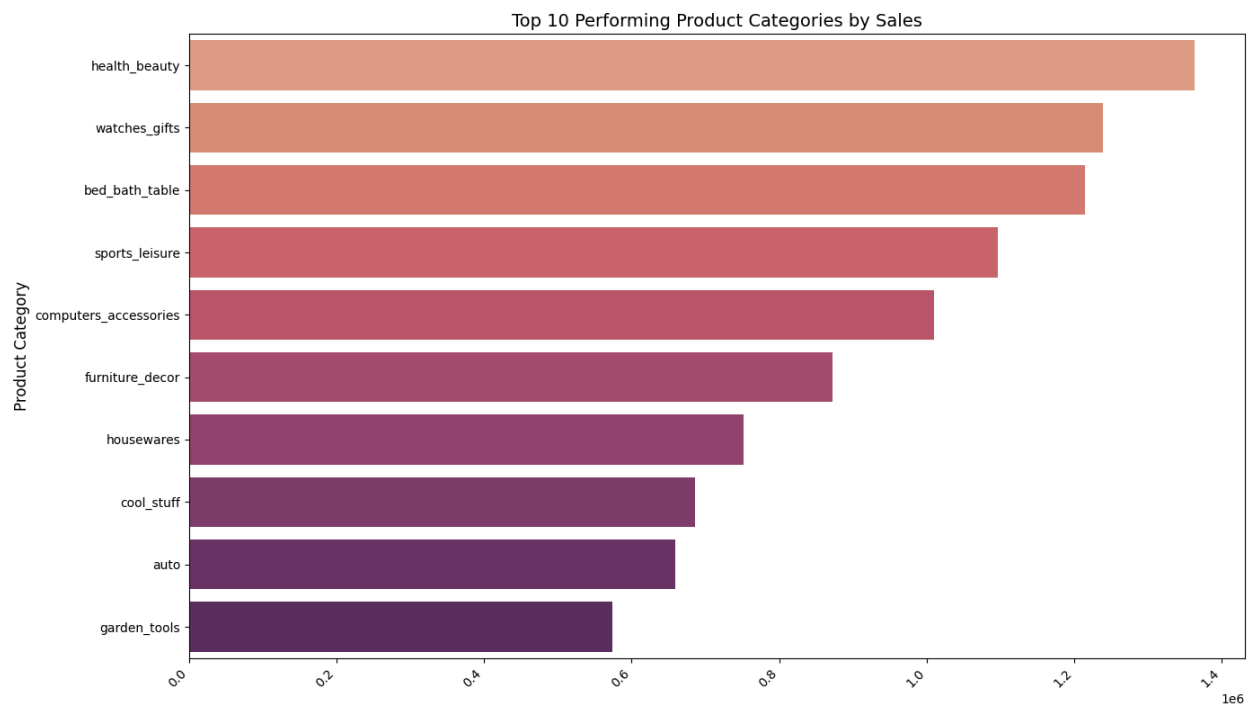
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Sravani Garapati (11936780)

Distribution of Total Amount spent by a Customer



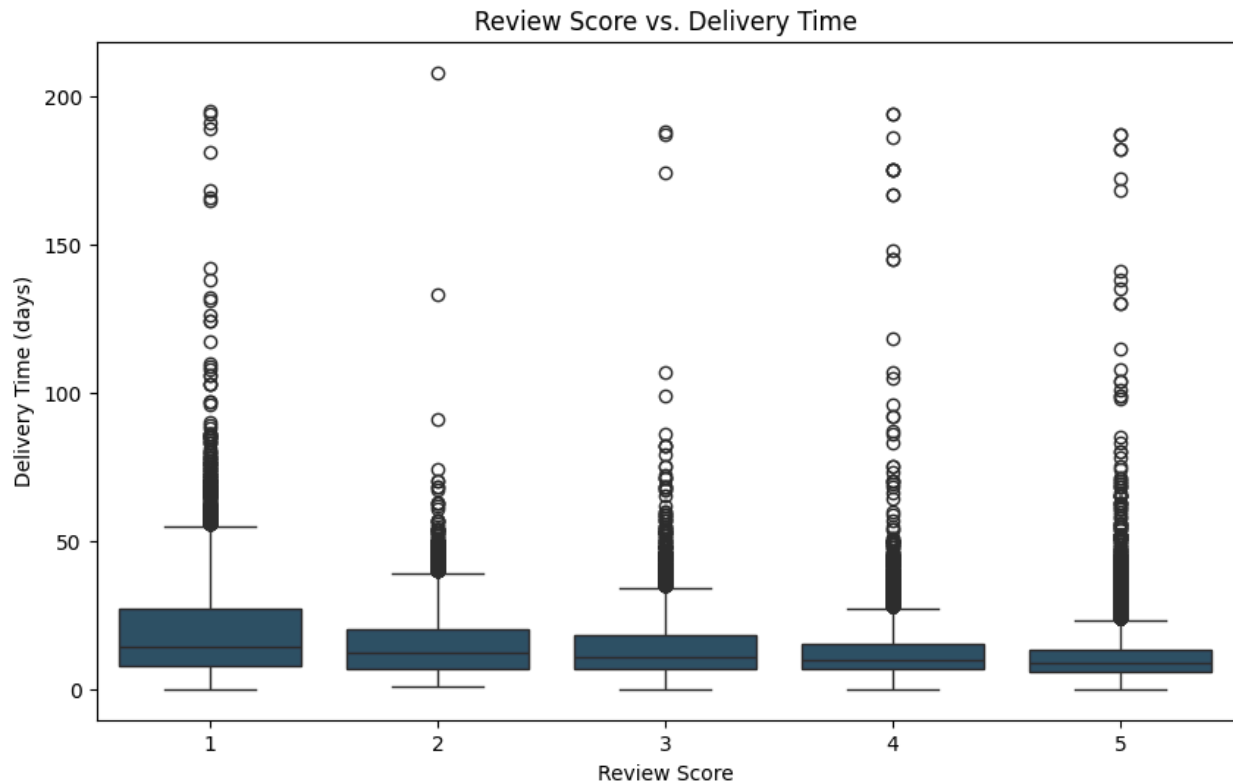
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Calculate Total Sales per Product Category



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Review Score VS Delivery Time



WORK DISTRIBUTION

Jaiharishan Arunagiri Veerakumar:

- Gathering Datasets and Importing the data in Google Colab Notebook.
- Outlier Analysis
- Descriptive Analysis
- Column Correlation
- Exploratory Data Analysis

Sravani Garapati:

- Dataset Analysis and Merging
- Dataset Cleaning
- Exploratory Data Analysis
- Report

Jaiharishan Arunagiri Veerakumar (62333614)
Sravani Garapati (11936780)

Next Steps

MILESTONE 2: (February 21, 2025 - March 21, 2025)

- Feature Engineering - MARCH 1
- Feature Selection - MARCH 8
- Data Modeling - March 12

MILESTONE 3: (March 24, 2025 - April 23, 2025)

- Evaluation and Interpretation - APRIL 6
- Interactive Dashboard - APRIL 14